

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Intermittent BD drum A150HAP10HP001 -7 Percent.pvdb

Date of Analysis : Sep 17, 2026 5:38pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 950.000 mm. ID
 Vessel Design Length, Tangent to Tangent 1575.00 mm.
 Distance of Bottom Tangent above Grade 300.00 mm.
 Specified Datum Line Distance 50.00 mm.
 Internal Design Temperature 150 °C
 Internal Design Pressure 3.500 bars
 External Design Temperature 150 °C
 External Design Pressure 1.034 bars
 Maximum Allowable Working Pressure 9.38 bars
 External Max. Allowable Working Pressure 4.29 bars
 Shop Test Pressure 4.550 bars
 Field Test Pressure 12.192 bars
 Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -48.0 °C
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Leg Baseplate		...	...		
Leg Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom D end	3.623	1.03	14.700	3.0000	No	No
Shell	3.606	1.03	16.900	3.0000	N/A	No
Top D end	3.500	1.03	14.800	3.0000	No	No

Liquid Level: 1337.50 mm. Dens.: 0.001 kg./cm3 Sp. Gr.: 0.936

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	0.0	50.0	12.0	9.0	4.5	5.5	0.85	1.00
Cylinder	1475.0	1475.0	10.0	10.0	4.5	7.0	0.85	1.00
Ellipse	1525.0	50.0	12.0	9.0	4.5	5.5	0.85	1.00

Local Stress Analysis Results:



Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
LEGS	WRC-107/537	0.174	n/a	Passed
LEGS	WRC-107/537	0.303	n/a	Passed
B2_3"	WRC-107/537	0.497	n/a	Passed
A2_4"	WRC-297	0.863	Shell	Passed
A1_2"	WRC-297	0.920	Nozzle	Passed
B1_8"	WRC-107/537	0.966	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	245. Kg
Total Earthquake Shear on top of all Legs	170. Kg
Total Wind Moment at top of all Legs	144. Kg-m.
Total Earthquake Moment at top of all Legs	138. Kg-m.
Wind Load Case Test Moment on Support	47. Kg-m.
Test Case Wind Shear on Support	81. Kg
Max. Wind Shear on one Leg (top & bottom)	98. Kg
Max. Earthq. Shear on one Leg (top & bottom)	68. Kg
Max. Wind Moment at base of one Leg	144. Kg-m.
Max. Earthquake Moment at base of one Leg	100. Kg-m.
Max. Vertical Load (Wt. + Wind) on one Leg	574. Kg
Max. Vertical Load (Wt. + Eq.) on one Leg	569. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	870.2 kg.
Shop Test - Fabricated + water (full)	2210.5 kg.
Shipping - Fab. + rem. intls.+ shipping app.	919.0 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	919.0 kg.
Empty - Fab. + intls. + details + wghts.	919.0 kg.
Operating - Empty + operating liquid (no ca)	1796.0 kg.
Field Test - Empty weight + test medium (full)	2259.3 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
B2_3"	14.6	...	...	OK	4.50	OK	No Calc[*]
L2_2"	16.9	...	...	OK	4.50	OK	No Calc[*]
L1_2"	16.9	...	...	OK	4.50	OK	No Calc[*]
A2_4"	16.9	OK	...	OK	4.50	OK	Passed
A2_3"	16.9	...	...	OK	4.50	OK	No Calc[*]
A1_2"	16.9	...	...	OK	4.50	OK	No Calc[*]
B1_8"	9.38	OK	...	OK	4.50	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 9.38 Nozzle : B18"
 Minimum MAWP Shells/Flanges : 9.38 Element : Bottom D end
 Minimum MAPnc Shells/Flanges : 22.2 Element : Bottom D end

Computed Vessel M.A.W.P. : 9.38 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	B2_3"	0.000	0.000	170.598
20	L2_2"	300.000	90.000	113.600
20	L1_2"	1050.000	0.000	113.600
20	A2_4"	850.000	180.000	201.662
20	A2_3"	850.000	180.000	163.130
20	A1_2"	850.000	90.000	163.130
30	B1_8"	0.000	0.000	323.658

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

*Diameter Limit clash between : A23" and A24".
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 182.40 mm..*

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
L2_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	221.14	150
L1_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	221.14	150
B2_3"	3.000 in	40	WNF	3.500	5.486	152.40	10.00	141.82	300
A2_3"	3.000 in	80	WNF	3.500	7.620	...	...	460.00	600
A1_2"	3.000 in	80	WNF	3.500	7.620	...	...	241.67	150
A2_4"	4.000 in	120	WNF	4.500	11.125	203.20	9.00	223.00	300
B1_8"	6.000 in	40	WNF	6.625	7.112	250.00	10.00	125.74	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
L2_2"	SA-105	9.000	10.000	...	...	...
L1_2"	SA-105	9.000	10.000	...	...	...
B2_3"	SA-106 B	9.000	9.000	8.000	10.000	...
A2_3"	SA-106 B	9.000	9.000	...	...	8.000
A1_2"	SA-105	10.000	10.000	...	...	8.000
A2_4"	SA-106 B	9.000	9.000	8.000	9.000	...
B1_8"	SA-106 B	9.000	10.000	8.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
L2_2"	250.000	90.0	209.55	0.00	Shell
L1_2"	1000.000	0.0	209.55	0.00	Shell
B2_3"	...	0.0	131.67	0.00	Bottom D end
A2_3"	800.000	180.0	350.00	100.00	Shell
A1_2"	800.000	90.0	131.67	100.00	Shell
A2_4"	800.000	180.0	209.55	0.00	Shell
B1_8"	...	0.0	112.62	0.00	Top D end

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm. THK X 950.0mm. ID X 50.0mm.	SA-516 70
2	INSULATION: 187mm X 30mm THK	HC
1	INSULATION: 1475mm X 30mm THK	HC
4	LEGS, SHAPE: L3X3X0.2500 1475.0mm	...
4	BASEPLATES: 154.0mm X 154.0mm X 13.0mm	SA-516 70
4	BASEPLATE BOLTS: 24.0mm	SA-36
4	LEG RE-PADS: 150.000mm X 150.000mm X 10.000mm	SA-516 70
1	CLASS 150 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 4.0" BLIND FLANGE(S)	SA-105
2	CYLINDRICAL SEGMENT 10.0mm. THK X 1187.5mm. X 1475.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 10.0mm. THK X 609.5mm. X 1475.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom D end	[10]	B	-29	-48	-29	0.544	9.000	0.85	No
Bottom D end	[7]	B	-23	-48	-29	0.367	12.000	0.85	No
Shell	[8]	B	-29	-48	-29	0.472	10.000	0.85	No
Top D end	[10]	B	-29	-48	-29	0.537	9.000	0.85	No
Top D end	[7]	B	-23	-48	-29	0.363	12.000	0.85	No
B23"	[1]	B	-8	-104	-29	0.076	4.801	1.00	No
Nozzle Flg	[5]	A	-18	-104					
L22"	[1]	A	-8	-104	-29	0.178	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
L12"	[1]	A	-8	-104	-29	0.174	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
A24"	[1]	B	-8	-104	-29	0.025	9.735	1.00	No
Nozzle Flg	[5]	A	-18	-104					
A23"	[1]	B	-8	-104	-29	0.036	6.668	1.00	No
Nozzle Flg	[5]	A	-18	-104					
A12"	[1]	A	-8	-104	-29	0.031	6.668	1.00	No
Nozzle Flg	[5]	A	-18	-104					
B18"	[1]	B	-8	-104	-29	0.077	6.223	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -8 -48

Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -48.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



Class From To : Basic Element Checks.

Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest) 3.5 bars
 Design Internal Temperature 150.0 °C
 Projection of Nozzle from Vessel Top 0 mm.
 Projection of Nozzle from Vessel Bottom 0 mm.
 Minimum Design Metal Temperature -10.0 °C
 Type of Construction Welded
 Special Service None
 Degree of Radiography RT-2
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE
 Load Case 13 IP+WE+EW
 Load Case 14 IP+WF+CW
 Load Case 15 IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	Km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm.
Distance Upwind of Crest	Lh	0 mm.
Distance from Crest to the Vessel	x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.
Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	2.000	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration	No	
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg
Design Pressure + Static Head	Y	
Consider MAP New and Cold in Noz. Design	N	
Consider External Loads for Nozzle Des.	Y	
Use ASME VIII-1 Appendix 1-9	N	
Perform Blast Load Analysis	No	
Material Database Year	Current w/Addenda or Code Year	

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/3
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Bottom D end
Distance "FROM" to "TO"	50 mm.
Inside Diameter	950 mm.
Element Thickness	9 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	150 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.52
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	232 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	-137.5 mm.
Height/Length of Liquid	187.5 mm.
Liquid Density	0.0009347 kg./cm3
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-137.5 mm.
Height/Length of Insulation	187.5 mm.
Thickness of Insulation	30 mm.
Density	0.0002 kg./cm3
Element From Node	10
Detail Type	Nozzle
Detail ID	B23"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	2/3
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	1475 mm.
Inside Diameter	950 mm.
Element Thickness	10 mm.



Internal Corrosion Allowance	3	mm.
Nominal Thickness	10	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	150	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.52	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1150	mm.
Liquid Density	0.0009347	kg./cm ³
Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1475	mm.
Thickness of Insulation	30	mm.
Density	0.0002	kg./cm ³
Element From Node	20	
Detail Type	Nozzle	
Detail ID	L22"	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	L12"	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	A24"	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	A23"	



Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	A12"	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Leg	
Detail ID	LEGS	
Dist. from "FROM" Node / Offset dist	150	mm.
Diameter at Leg Centerline	1039.2	mm.
Leg Orientation	3	
Number of Legs	4	
Section Identifier	L3X3X0.2500	
Length of Legs	1475	mm.

Element From Node	20	
Detail Type	Weight	
Detail ID	Lifting WT	
Dist. from "FROM" Node / Offset dist	1250	mm.
Miscellaneous Weight	25	Kgf
Offset from Element Centerline	0	mm.

Element From Node	20	
Detail Type	Weight	
Detail ID	insu. CLIP WT	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	35	Kgf
Offset from Element Centerline	0	mm.

Element#	3/3	
Element From Node	30	
Element To Node	40	
Element Type	Elliptical	
Description	Top D end	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	950	mm.
Element Thickness	9	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	150	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.52	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	187.5 mm.
Thickness of Insulation	30 mm.
Density	0.0002 kg./cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	B18"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	40
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Bottom D end		...	50	...	50
Shell		...	1525	...	1475
Top D end		...	1575	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Bottom D end		3.6227	12	3	950	117.3	1
Shell		3.6055	10	3	950	117.3	2
Top D end		3.5	12	3	950	117.3	3

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Bottom D end		3.5	14.7	22.2	9	4.5	1
Shell		3.5	16.9	24.4	10	4.5	2
Top D end		3.5	14.8	22.2	9	4.5	3

Minimum 9.38 19.6

Warning: A standard flange is governing the MAP(NC)!

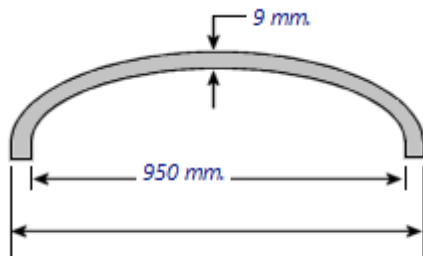
MAWP: 9.38 bars, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom D end

Elliptical Head Design Information:					
Design Pressure	P	3.623 bars	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		150 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	950.000 mm.	Actual Stress		33.7 N./mm^2
Spec. Min. Thk t,e, or sa		9.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		11813.5 cm^2	SG of Contents		0.936
Element Volume		147697.7 cm3	Weight of Contents		114.2 Kg
Empty Weight		117.1 Kg	Operating Weight		231.3 Kg
			Comp. All. Ope. Stress		116.9 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.938 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.62 \cdot 956 \cdot 0.99) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 3.62)$$

$$= 1.4646 + 3.0000 = 4.4646 \text{ mm.}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.123 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 6) / (0.99 \cdot 956 + 0.2 \cdot 6)$$

$$= 14.8 - 0.12 = 14.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 9) / (1 \cdot 950 + 0.2 \cdot 9)$$

$$= 22.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$



$$= (3.62 \cdot (0.99 \cdot 956 + 0.2 \cdot 6)) / (2 \cdot 0.85 \cdot 6)$$

$$= 33.717 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (3.62 \cdot 478) / (138 \cdot 0.85 - 0.6 \cdot 3.62) + 3$$

$$= 4.479 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.123 bars](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 9) / (478 + 0.6 \cdot 9)$$

$$= 21.8 - 0.12 = 21.7 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (956 / (2 \cdot 241))^2) / 6$$

$$= 0.991710$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.373 %
[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 9, tr = 3.84, c = 3 mm., E* = 0.85
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.54$, Temp. Reduction = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12, tr = 3.89, c = 3 mm., E* = 0.85
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.37$, Temp. Reduction = 65 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.501 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	9.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.544	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.938 mm.

EXTERNAL PRESSURE RESULTS for: Bottom D end

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199807168 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.000	968.00	161.33	0.0008609	80.42



$$MAEP = B/(K0*Do/t) = 80.4/(0.9 *161) = 5.54 \text{ bars}$$

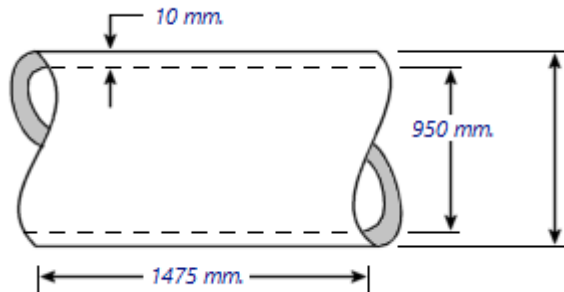
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.507	968.00	386.08	0.0003597	35.94

$$MAEP = B/(K0*Do/t) = 35.9/(0.9 *386) = 1.03 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	3.606 bars	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		150 °C
Length		1475.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	950.000 mm.	Actual Stress		29.2 N./mm^2
Spec. Min. Thk t _e , or s _a		10.0000 mm.	Spec. Nominal Thk.		10.000 mm.
Surface Area		44948.3 cm^2	SG of Contents		0.936
Element Volume		1045699.6 cm3	Weight of Contents		762.8 Kg
Empty Weight		344.8 Kg	Operating Weight		1107.6 Kg
			Comp. All. Ope. Stress		121.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.61 \cdot 478) / (138 \cdot 0.85 - 0.6 \cdot 3.61)$$

$$= 1.4721 + 3.0000 = 4.4721 \text{ mm.}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.106 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 7) / (478 + 0.6 \cdot 7)$$

$$= 17 - 0.11 = 16.9 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 10) / (475 + 0.6 \cdot 10)$$

$$= 24.4 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.61 \cdot (478 + 0.6 \cdot 7)) / (0.85 \cdot 7)$$

$$= 29.222 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.042 %



Minimum Design Metal Temperature Results:

Govrn. thk, tg = 10, tr = 3.88, c = 3 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.47, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.484 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.472	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199807168 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.000	970.00	1733.33	138.57	1.7869	0.0004463	44.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*44.6)/(3*139) = 4.29 \text{ bars}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.964	970.00	1733.33	244.72	1.7869	0.0001900	18.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*19)/(3*245) = 1.03 \text{ bars}$$

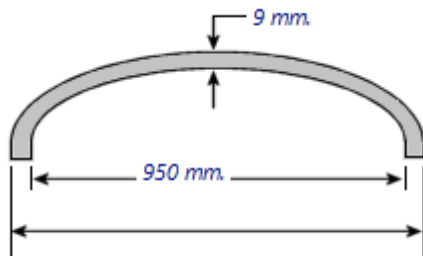
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.000	970.00	6531.45	138.57	6.7335	0.0001076	10.75

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.8)/(3*139) = 1.03 \text{ bars}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Top D end

Elliptical Head Design Information:					
Design Pressure	P	3.500 bars	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		150 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	950.000 mm.	Actual Stress		32.6 N./mm^2
Spec. Min. Thk t,e, or sa		9.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		11813.5 cm^2	SG of Contents		0.000
Element Volume		147697.7 cm3	Weight of Contents		0.0 Kgf
Empty Weight		117.1 Kgf	Operating Weight		117.1 Kgf
			Comp. All. Ope. Stress		116.9 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.938 mm.

Required Thickness due to Internal Pressure [t_r]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.5 \cdot 956 \cdot 0.99) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 3.5)$$

$$= 1.4150 + 3.0000 = 4.4150 \text{ mm.}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 6) / (0.99 \cdot 956 + 0.2 \cdot 6)$$

$$= 14.8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 9) / (1 \cdot 950 + 0.2 \cdot 9)$$

$$= 22.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [S_{act}]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.5 \cdot (0.99 \cdot 956 + 0.2 \cdot 6)) / (2 \cdot 0.85 \cdot 6)$$

= 32.575 N./mm²

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (3.5 \cdot 478) / (138 \cdot 0.85 - 0.6 \cdot 3.5) + 3$
 $= 4.429$ mm.

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 9) / (478 + 0.6 \cdot 9)$
 $= 21.8$ bars

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$
 $= (2 + (956 / (2 \cdot 241)))^2 / 6$
 $= 0.991710$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.373 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 9, tr = 3.79, c = 3 mm., E* = 0.85
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.54$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12, tr = 3.84, c = 3 mm., E* = 0.85
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.36$, Temp. Reduction = 68 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.378 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	9.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.537	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.938 mm.

EXTERNAL PRESSURE RESULTS for: Top D end

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199807168 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.000	968.00	161.33	0.0008609	80.42

MAEP = $B / (K0 \cdot Do / t) = 80.4 / (0.9 \cdot 161) = 5.54$ bars



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.507	968.00	386.08	0.0003597	35.94

$$MAEP = B / (K0 * Do / t) = 35.9 / (0.9 * 386) = 1.03 \text{ bars}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	12.192 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.480 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	10.316 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	9.378 bars

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 3.5 • 1
 = 4.550 bars

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	28.9	234.0	0.123	4.64
Shell	26.3	234.0	0.112	4.64
Top D end	28.9	234.0	0.123	4.64

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2_3"	Nozzle	118.00	118.00	1.000
B2_3"	Pad	138.00	138.00	1.000
L2_2"	Nozzle	138.00	138.00	1.000
L1_2"	Nozzle	138.00	138.00	1.000
A2_4"	Nozzle	118.00	118.00	1.000
A2_4"	Pad	138.00	138.00	1.000
A2_3"	Nozzle	118.00	118.00	1.000
A1_2"	Nozzle	138.00	138.00	1.000
B1_8"	Nozzle	118.00	118.00	1.000
B1_8"	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
B2_3"	4.11	216.00	0.019
L2_2"	1.15	225.00	0.005
L1_2"	1.15	225.00	0.005
A2_4"	2.54	216.00	0.012
A2_3"	2.91	216.00	0.013
A1_2"	2.91	225.00	0.013
B1_8"	6.09	216.00	0.028



Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	12.192 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.480 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	10.316 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	9.378 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 9.38 • 1
 = 12.192 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	115.3	234.0	0.493	12.39
Shell	100.2	234.0	0.428	12.36
Top D end	113.7	234.0	0.486	12.22

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2_3"	Nozzle	118.00	118.00	1.000
B2_3"	Pad	138.00	138.00	1.000
L2_2"	Nozzle	138.00	138.00	1.000
L1_2"	Nozzle	138.00	138.00	1.000
A2_4"	Nozzle	118.00	118.00	1.000
A2_4"	Pad	138.00	138.00	1.000
A2_3"	Nozzle	118.00	118.00	1.000
A1_2"	Nozzle	138.00	138.00	1.000
B1_8"	Nozzle	118.00	118.00	1.000
B1_8"	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
B2_3"	10.98	216.00	0.051
L2_2"	3.07	225.00	0.014
L1_2"	3.07	225.00	0.014
A2_4"	6.76	216.00	0.031
A2_3"	7.75	216.00	0.036
A1_2"	7.75	225.00	0.034
B1_8"	16.03	216.00	0.074



Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	968	6	0.00086088	80.4214
20	30	1733.33	970	7	0.0004463	44.5898
30	40	No Calc	968	6	0.00086088	80.4214

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	9	5.50723	1.03419	5.54
20	30	10	6.96366	1.03419	4.29
30	40	9	5.50723	1.03419	5.54

Minimum

4.29

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	1733.33	6531.45	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	117.07	147698	87.8022	150287	8.19487
20	30	344.782	1045699	242.101	1058950	24.1347
30	40	117.07	147698	87.8022	150287	8.19487
Total		578	1341095.00	417	1359524.75	40

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Liqd	114.181	...	-93.75	...	Liquid 10
10	Insl	10.4916	...	-43.75	...	Ins: 10
10	Nozl	11.1879	...	-348.837	...	B2_3"
20	Liqd	762.808	...	575	...	Liquid: 20
20	Insl	27.8356	...	737.5	...	Ins: 20
20	Nozl	10.9926	...	250	-579.775	L2_2"
20	Nozl	8.34136	579.775	1000	...	L1_2"
20	Nozl	33.8733	-579.775	800	...	A2_4"
20	Nozl	16.5013	-650.006	800	...	A2_3"
20	Nozl	9.44194	...	800	-540.837	A1_2"
20	Legs	55.7933	...	-587.5	...	LEGS
20	Wght	25	...	1250	...	Lifting WT
20	Wght	35	...	...	...	insu. CLIP WT
30	Insl	10.4916	...	93.75	...	Ins: 30
30	Nozl	44.6521	...	339.312	...	B1_8"

Total Weight of Each Detail Type:

Liquid	877.0
Insulation	48.8
Nozzles	135.0
Legs	55.8
Weights	60.0

Sum of the Detail Weights 1176.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	619.4	619.4	619.4	619.4	619.4	619.4
Nozzles	135.0	135.0	135.0	135.0	135.0	135.0
Legs	55.8	55.8	55.8	55.8	55.8	55.8
Wld Weights	60.0	60.0	60.0	60.0	60.0	60.0
Insulation	...	...	48.8	48.8	48.8	48.8
Ope. Liquid	...	...	...	...	...	877.0
Test Liquid	...	1340.3	...	...	...	...
Totals	870.2	2210.5	919.0	919.0	919.0	1796.0

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:



Fabricated Wt.	- Bare weight without removable internals	870.2 kg.
Shop Test Wt.	- Fabricated weight + water (full)	2210.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	919.0 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	919.0 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	919.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1796.0 kg.
Field Test Wt.	- Empty weight + test medium (full)	2259.3 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	366.0 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	702.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	333.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	11813.5	11275.2
20	30	44948.3	44021.6
30	40	11813.5	11275.2
Total		68575.3	66572.0 cm2
Total		73.8	71.7 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	146.944	261.125	294.551	...	146.944
20	Legs	54.4986	132.072	160.776	25.5292	54.4986
Legs	30	481.404	1166.64	1420.19	...	481.404
30	40	180.408	180.408	328.016	...	180.408

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	Legs	-146.944	-261.125	-294.551
Legs	30	661.812	1347.05	1748.2
30	40	180.408	180.408	328.016

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	Legs	25.5292	25.5292	25.5292
Legs	30	...	...	...
30	40	...	...	...

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
B2_3"	45.10	51.00	150	300	GR 1.1	48.759	18.891	...	18.891
L2_2"	15.80	19.60	150	150	GR 1.1	...	...	...	...
L1_2"	15.80	19.60	150	150	GR 1.1	...	...	...	...
A2_4"	45.10	51.00	150	300	GR 1.1	44.781	22.869	45.100	22.869
A2_3"	90.10	102.00	150	600	GR 1.1	...	...	...	...
A1_2"	15.80	19.60	150	150	GR 1.1	22.771	11.990	15.800	0.930
B1_8"	90.10	102.00	150	600	GR 1.1	64.665	70.487	25.436	70.486
Min Rating	11.990	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
B2_3"	111.125	0.50	120	489	9.378	18.891	Pass
A2_4"	138.176	0.50	214	653	9.378	22.869	Pass
A1_2"	111.125	1.20	54	326	9.378	11.990	Pass
B1_8"	192.151	0.50	856	1305	9.378	70.487	Pass

Evaluating Flange on Nozzle [B23"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(120) + 4 \cdot \text{abs}(489)111.125 \leq \pi \cdot 111.125^3 \cdot ((45.100 - 9.378) + .500 \cdot 45.100)$$

$$= 0.21436E+04 \leq 0.25618E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B23" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.100, 0.500 \cdot 45.100 + 45.100 - 2143.6 / (\pi \cdot 111.125^3))$$

$$= \min(45.100, 18.891)$$

$$= 18.891 \text{ bars}$$

Evaluating Flange on Nozzle [A24"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(214) + 4 \cdot \text{abs}(653)138.176 \leq \pi \cdot 138.176^3 \cdot ((45.100 - 9.378) + .500 \cdot 45.100)$$

$$= 0.37848E+04 \leq 0.49251E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A24" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.100, 0.500 \cdot 45.100 + 45.100 - 3784.8 / (\pi \cdot 138.176^3))$$

$$= \min(45.100, 22.869)$$

$$= 22.869 \text{ bars}$$

Evaluating Flange on Nozzle [A12"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(54) + 4 \cdot \text{abs}(326)111.125 \leq \pi \cdot 111.125^3 \cdot ((15.800 - 9.378) + 1.200 \cdot 15.800)$$

$$= 0.10011E+04 \leq 0.11159E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A12" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(15.800, 1.200 \cdot 15.800 + 15.800 - 1001.1 / (\pi \cdot 111.125^3))$$

$$= \min(15.800, 11.990)$$



= 11.990 bars

Evaluating Flange on Nozzle [B18"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(856) + 4 \cdot \text{abs}(1305) 192.151 \leq \pi \cdot 192.151^3 [(90.101 - 9.378) + .500 \cdot 90.101] \\ &= 0.14698E+05 \leq 0.28587E+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle B18" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(90.101, 0.500 \cdot 90.101 + 90.101 - 14697.7 / (\pi \cdot 192.151^3)) \\ &= \min(90.101, 70.487) \\ &= 70.487 \text{ bars} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 25.5184 Hz.

The Natural Frequency for the Vessel (Ope...) is 18.5486 Hz.

The Natural Frequency for the Vessel (Filled) is 16.0976 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	18.549 Hz
Natural Frequency of Vessel (Empty)	25.518 Hz
Natural Frequency of Vessel (Test)	16.098 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.514
Structure Height to Diameter ratio	1.858
Height to top of Structure	1851.500 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Zg = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 186 + 300 = 486$ mm.
 $= 1.59$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (1.59 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 589788)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [162 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.51 \cdot 4.49$$

$$= 65.1 \text{ lbs. } [29.5 \text{ Kg}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Bottom D end	485.8	0.000	0.000	0.000	1.035	1.000	161.849
Shell	1325.0	0.000	0.000	0.000	1.035	1.000	161.849
Top D end	2203.2	0.000	0.000	0.000	1.035	1.000	161.849

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of (D/L²)(104)
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kg
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/1575 = 0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)
 :: [3.38/5.17²] • 10⁴ = 1265 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.40047E-06 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 1624 / (1575 \cdot 1010^2)$$

$$= 0.10108E-05$$

Since Vp is > 0.40047E-06 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kg
----	------------------------------	------------------	--------------------------------	-------------



Lifting WT	0.00	1600.00	161.85	0.00
insu. CLIP WT	0.00	350.00	161.85	0.00

The Natural Frequency for the Vessel (Ope...) is 18.5486 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	485.791	1562.56	4174.58	161.849	32.4899
20	30	1325	1565.6	23092.6	161.849	179.725
30	40	2203.21	1562.56	4174.58	161.849	32.4899

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor		1.500
Force Modification Factor R		2.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		No
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(10.10.75)$
 $= 0.114$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/18.5)$
 $= 0.054$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.11, 0.054)$ per 12.8.2
 $= 0.054$

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.22 \cdot 1740 \cdot 1.5$
 $= 170.196$ Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 18.5486 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	25	261.125	0.84248	0.43719	...
20	Legs	200	132.072	3.40888	1.29715	...



Legs	30	862.5	1166.64	129.857	49.4135	...
30	40	1550	180.408	36.0876	33.2785	...

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	290.709	...	...	...	...
20	Legs	75	32.4899	0.84248	4.57172	0.11855
Legs	30	662.5	244.705	170.196	143.767	138.426
30	40	1465.71	32.4899	36.0876	4.57172	5.07795

Note:
 The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:
 Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	...	6.11295	6.11295	0.0062161
20	Legs	32.4899	6.11296	6.11299	0.0062166
Legs	30	244.705	6.11394	6.11549	0.0062186
30	40	32.4899	6.11555	6.11561	0.0062186

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	520.294	3251.84
20	30	521.306	3258.16
30	40	520.294	3251.84

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 7.875 Actual deflection : 6.116 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	271.15	181.332	6440959	4334171
20	30	301.594	211.775	7164399	5061955
30	40	271.15	181.332	6440959	4334171

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	165.597	280.795	-116.886	-117.151
20	Legs	165.597	280.795	-121.365	-121.648
Legs	30	165.597	280.795	-121.365	-121.648
30	40	165.597	280.795	-116.886	-117.151

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	13.8725	-4.19752	48.323
20	30	11.8807	-3.60898	41.3849
30	40	13.8725	-4.19752	48.323

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.068047	0.12092	0.1364	0.049458	0.049458
Legs	30	-0.30647	-0.27055	-0.30647	...	...
30	40	-0.097569	-0.097569	-0.097569	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.049458	0.0088569	0.00022966	0.0029228	...
Legs	30	...	0.27852	0.26818	0.091913	...
30	40	...	0.010344	0.01149	0.0034136	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	0.00011918
Legs	30	...	...	...	0.15749
30	40	...	...	...	0.010595

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	...
20	Legs	...	...
Legs	30	...	...
30	40	...	...

Longitudinal Stresses due to User Forces and Moments:



From	To	Wind For/Mom Corroded N./mm^2	Earthquake For/Mom Corroded N./mm^2	Wind For/Mom No Corrosion N./mm^2	Earthquake For/Mom No Corrosion N./mm^2
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm^2
10	20	...
20	Legs	...
Legs	30	...
30	40	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	116.89	0.0000	-0.0000
20	0.13	165.60		121.36	0.0008	
20		165.60	-0.58	121.36		0.0048
30		165.60	-0.11	116.89		0.0009

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	116.89	0.0000	-0.0000
20	0.12	165.60		121.36	0.0007	
20		165.60	-0.46	121.36		0.0038
30		165.60	-0.11	116.89		0.0009

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	116.89	0.0000	-0.0000
20	0.18	165.60		121.36	0.0011	
20	0.01	165.60	-0.55	121.36	0.0000	0.0045
30		165.60	-0.11	116.89		0.0009

Analysis of Load Case 4 : NP+OW+EQ+FS+BS



From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	116.89	0.0000	-0.0000
20	0.17	165.60		121.36	0.0010	
20		165.60	-0.54	121.36		0.0044
30		165.60	-0.11	116.89		0.0009

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.79	0.00	117.15	0.0000	-0.0000
20	0.19	280.79		121.65	0.0007	
20		280.79	-0.40	121.65		0.0033
30		280.79	-0.10	117.15		0.0009

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.79	0.00	117.15	0.0000	-0.0000
20	0.19	280.79		121.65	0.0007	
20		280.79	-0.31	121.65		0.0025
30		280.79	-0.10	117.15		0.0008

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.87	165.60		116.89	0.0838	
20	12.06	165.60		121.36	0.0728	
20	11.89	165.60		121.36	0.0718	
30	13.79	165.60		116.89	0.0832	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.87	165.60		116.89	0.0838	
20	12.05	165.60		121.36	0.0728	
20	11.88	165.60		121.36	0.0717	
30	13.79	165.60		116.89	0.0833	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-4.20	116.89		0.0359
20		165.60	-3.55	121.36		0.0292
20		165.60	-4.16	121.36		0.0343
30		165.60	-4.31	116.89		0.0368

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-4.20	116.89		0.0359
20		165.60	-3.54	121.36		0.0291
20		165.60	-4.15	121.36		0.0342
30		165.60	-4.31	116.89		0.0368

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	48.32	280.79		117.15	0.1721	
20	41.57	280.79		121.65	0.1481	
20	41.17	280.79		121.65	0.1466	
30	48.23	280.79		117.15	0.1718	



Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	48.32	280.79		117.15	0.1721	
20	41.57	280.79		121.65	0.1480	
20	41.08	280.79		121.65	0.1463	
30	48.23	280.79		117.15	0.1717	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.87	165.60		116.89	0.0838	
20	12.00	165.60		121.36	0.0725	
20	11.57	165.60		121.36	0.0699	
30	13.77	165.60		116.89	0.0832	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.87	165.60		116.89	0.0838	
20	11.97	165.60		121.36	0.0723	
20	11.69	165.60		121.36	0.0706	
30	13.81	165.60		116.89	0.0834	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.87	165.60		116.89	0.0838	
20	12.05	165.60		121.36	0.0728	
20	11.61	165.60		121.36	0.0701	
30	13.77	165.60		116.89	0.0832	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.87	165.60		116.89	0.0838	
20	12.00	165.60		121.36	0.0725	
20	11.57	165.60		121.36	0.0699	
30	13.77	165.60		116.89	0.0832	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	116.89	0.0000	-0.0000
20	0.17	165.60		121.36	0.0010	
20		165.60	-0.27	121.36		0.0022
30		165.60	-0.10	116.89		0.0008

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.87	138.00		97.41	0.1005	
20	12.05	138.00		101.14	0.0873	
20	11.61	138.00		101.14	0.0841	
30	13.77	138.00		97.41	0.0998	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-4.20	97.41		0.0431
20		138.00	-3.54	101.14		0.0350



20	138.00	-3.88	101.14	0.0384
30	138.00	-4.30	97.41	0.0441

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.87	165.60		116.89	0.0838	
20	12.05	165.60		121.36	0.0728	
20	11.61	165.60		121.36	0.0701	
30	13.77	165.60		116.89	0.0832	

Absolute Maximum of the all of the Stress Ratio's 0.1721

Governing Element: Bottom D end
 Governing Load Case 11 : HP+HW+HI

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom D end	48.3	...	...	...	48.3	+ Tens
Bottom D end	48.3	...	...	...	...	- Comp
Shell	41.4	0.1	...	...	41.6	+ Tens
Shell	41.4	0.1	...	...	...	- Comp
Shell	41.4	-0.3	...	0.1	41.2	+ Tens
Shell	41.4	-0.3	...	-0.1	...	- Comp
Top D end	48.3	-0.1	...	...	48.2	+ Tens
Top D end	48.3	-0.1	...	...	...	- Comp



Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	531.421 mm.
Center of Gravity of the Insulation	787.500 mm.
Center of Gravity of the Nozzles	1053.726 mm.
Center of Gravity of the Legs	-537.500 mm.
Center of Gravity of the Added Weights (Operating)	570.833 mm.
Center of Gravity of the Added Weights (Empty)	570.833 mm.
Center of Gravity of Bare Shell New and Cold	787.500 mm.
Center of Gravity of Bare Shell Corroded	787.500 mm.
Vessel CG in the Operating Condition	617.768 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	732.021 mm.
Vessel CG in the Test Condition	764.932 mm.

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle L3X3X0.2500

USA AISC 1989 Steel Table

Overall Leg Length		1475.000	mm.
Effective Leg Length	Leglen	1245.000	mm.
Distance Leg Up Side of Vessel		150.000	mm.
Number of Legs	Nleg	4	
Cross Sectional Area for L3X3X0.2500	Aleg	9.290	cm ²
Section Inertia (strong axis)		51.613	cm ⁴
Section Inertia (weak axis)		51.613	cm ⁴
Section Modulus (strong axis)		9455.335	mm. ³
Section Modulus (weak axis)		9455.335	mm. ³
Radius of Gyration (strong axis)		23.622	mm.
Radius of Gyration (weak axis)		23.622	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		143.8	Kg-m.
Total Weight Load at top of Legs	W	1743.9	Kgf
Total Shear force at top of Legs		244.7	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	150.000	mm.
Pad Length along Vessel Axis	C22P	150.000	mm.
Pad Thickness	Tpad	10.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (245) (82.6/207)$
 $= 97.55 \text{ Kg}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (17102/4 + (1409762 / (2 \cdot 520))) / 929$
 $= 6.06 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (1744/4) / 9.29$
 $= 4.60 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	76.200	76.200	6.3500
Distance to geometric centroid:	21.387	21.387	
Arm about YY:	18.212	19.888	
Arm about ZZ:	16.713	18.212	
Leg areas:	4.8387	4.4355	
Geometric inertia components YY:	16.211	35.578	
Geometric inertia components ZZ:	36.929	14.860	
Geometric inertias Iy & Iz:	51.789	51.789	
Product of inertia:	30.793		
Mohrs Radius:	30.793		
Average Inertia:	51.789		

QFACT =	1.0000	FBZ	=	0.14893
Principal Axis Inertias (Z&W)			=	20.996 82.582
Angle to Principal Axis			=	45.000



Distances to extreme fibers CW & CZ = 53.882 23.636
 FOB from Eq 5-5 = 0.99348
 Bending allowables Fby & Fbz = 0.16383 0.14893

Shear Center Coordinates Wo & Zo: 25.368 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.30000E-01 2.7256
 AREA, LENGTH, Kw, Kz: 1.4400 49.016 1.0000 1.0000
 H, Few, Fez, Fej: 0.63403 166.97 42.451 86.726
 Fe computed from C4-1: 68.965

Initial (Kl/r)_{max}, & (Kl/r)_{equiv} = 82.816 64.975
 Final (Kl/r)_{max}, & Cc = 82.816 127.18
 Fa based on Eq 4-1 = 0.10424

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	94.81	198.53
Strong Axis Bending :	54.95	218.38
Axial Compression :	6.06	138.96

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.773

AISC Unity Check : 0.773, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Ope Condition

Rectangular Attachment Parameter C11 107.763 mm.
 Rectangular Attachment Parameter C22 130.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	950.000	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	150.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Rectangular
Parameter C11	C11	107.76 mm.
Parameter C22	C22	130.95 mm.
Thickness of Reinforcing Pad	Tpad	10.000 mm.
Pad Parameter C11P	C11p	150.000 mm.
Pad Parameter C22P	C22p	150.000 mm.
Design Internal Pressure	Dp	3.500 bars
Include Pressure Thrust		No
Vessel Centerline Direction Cosine	Vx	0.000
Vessel Centerline Direction Cosine	Vy	1.000
Vessel Centerline Direction Cosine	Vz	0.000
Nozzle Centerline Direction Cosine	Nx	1.000
Nozzle Centerline Direction Cosine	Ny	0.000
Nozzle Centerline Direction Cosine	Nz	0.000

Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	436.0	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	15.1	Kg-m.

Internal Pressure (SUS)	P	3.50	bars
Include Pressure Thrust		No	

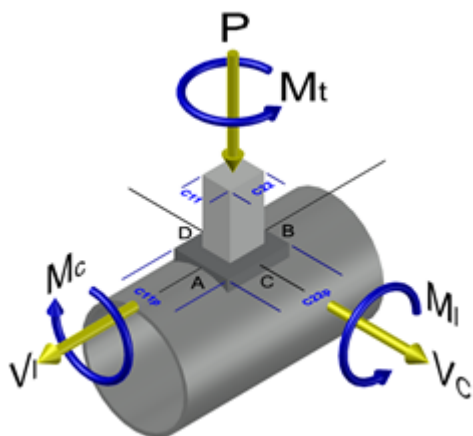
Global Force (OCC)	Fx	97.6	Kgf
Global Force (OCC)	Fy	138.3	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	65.5	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	0.00	bars
------------------------------------	------	------	------

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	436.0	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-15.1	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 28.62

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.136	4C	4.819	(A,B)
N(PHI) / (P/Rm)	0.136	3C	3.938	(C,D)



M(PHI) / (P)	0.119	2C1	0.088	(A,B)
M(PHI) / (P)	0.119	1C	0.123	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.118	3A	0.873	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.131	1A	0.094	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.126	3B	2.990	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.127	1B	0.043	(A,B,C,D)
N(x) / (P/Rm)	0.128	3C	4.077	(A,B)
N(x) / (P/Rm)	0.128	4C	4.906	(C,D)
M(x) / (P)	0.129	1C1	0.118	(A,B)
M(x) / (P)	0.129	2C	0.080	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.118	4A	1.280	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.149	2A	0.048	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.126	4B	0.876	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.135	2B	0.068	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.8	0.8	-0.8	-0.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.2	-2.2	-2.2	2.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.9	-1.4	-2.9	1.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		3.2	-3.2	-3.2	3.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		3.4	-2.9	-3.4	2.9	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Str. Int.		3.4	2.9	3.4	2.9	1.9	1.9	1.9	1.9

Dimensionless Parameters used : Gamma = 68.79

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.156	4C	9.591	(A,B)
N(PHI) / (P/Rm)	0.156	3C	6.193	(C,D)
M(PHI) / (P)	0.156	2C1	0.036	(A,B)
M(PHI) / (P)	0.156	1C !	0.070	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.156	3A	2.738	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.156	1A	0.074	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.156	3B	6.999	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.156	1B	0.025	(A,B,C,D)
N(x) / (P/Rm)	0.156	3C	6.193	(A,B)
N(x) / (P/Rm)	0.156	4C	9.591	(C,D)
M(x) / (P)	0.156	1C1	0.070	(A,B)
M(x) / (P)	0.156	2C !	0.038	(C,D)



N(x) / (MC/(Rm**2 * Beta)) 0.156 4A 5.228 (A,B,C,D)
 M(x) / (MC/(Rm * Beta)) 0.156 2A 0.036 (A,B,C,D)
 N(x) / (ML/(Rm**2 * Beta)) 0.156 4B 2.760 (A,B,C,D)
 M(x) / (ML/(Rm * Beta)) 0.156 2B 0.032 (A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		4.1	4.1	-4.1	-4.1	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		6.0	-6.0	-6.0	6.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		10.1	-1.9	-10.1	1.9	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.6	1.6	-1.6	-1.6	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		7.8	-7.8	-7.8	7.8	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		9.4	-6.2	-9.4	6.2	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Str. Int.		10.1	6.2	10.1	6.2	4.1	4.1	4.1	4.1

WRC 107 Stress Calculation for OCCasional loads:

Radial Load P 97.6 Kg
 Circumferential Shear VC -0.0 Kg
 Longitudinal Shear VL 138.3 Kg
 Circumferential Moment MC -0.0 Kg-m.
 Longitudinal Moment ML -65.5 Kg-m.
 Torsional Moment MT -0.0 Kg-m.

Dimensionless Parameters used : Gamma = 28.62
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.6	-0.6	-0.6	-0.6	-0.5	-0.5	-0.5	-0.5
Circ. Bend. P		-1.7	1.7	-1.7	1.7	-2.4	2.4	-2.4	2.4
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		3.4	3.4	-3.4	-3.4	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		9.4	-9.4	-9.4	9.4	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		10.4	-4.8	-15.0	7.2	-2.9	2.0	-2.9	2.0
Long. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.6	-0.6	-0.6	-0.6



Long. Bend. P	-2.3	2.3	-2.3	2.3	-1.6	1.6	-1.6	1.6
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML	1.1	1.1	-1.1	-1.1	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	13.7	-13.7	-13.7	13.7	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	12.0	-10.8	-17.7	14.5	-2.2	1.0	-2.2	1.0
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Str. Int.	12.0	10.8	17.7	14.5	3.0	2.1	3.0	2.1

Dimensionless Parameters used : Gamma = 68.79
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-2.7	-2.7	-2.7	-2.7	-1.8	-1.8	-1.8	-1.8
Circ. Bend. P		-4.2	4.2	-4.2	4.2	-8.2	8.2	-8.2	8.2
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		17.8	17.8	-17.8	-17.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		26.2	-26.2	-26.2	26.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		37.0	-6.9	-50.9	9.9	-10.0	6.5	-10.0	6.5
Long. Memb. P		-1.8	-1.8	-1.8	-1.8	-2.7	-2.7	-2.7	-2.7
Long. Bend. P		-8.2	8.2	-8.2	8.2	-4.4	4.4	-4.4	4.4
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		7.0	7.0	-7.0	-7.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		34.0	-34.0	-34.0	34.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		31.1	-20.6	-51.0	33.4	-7.1	1.7	-7.1	1.7
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Str. Int.		37.0	20.6	51.0	33.4	10.1	6.6	10.1	6.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		9.7	10.0	9.7	10.0	9.7	10.0	9.7	10.0
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		9.7	10.0	9.7	10.0	9.7	10.0	9.7	10.0
Circ. Pl (SUS)		0.8	0.8	-0.8	-0.8	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		2.8	2.8	-3.9	-3.9	-0.5	-0.5	-0.5	-0.5
Circ. Pl(TOTAL)		3.6	3.6	-4.7	-4.7	-0.5	-0.5	-0.5	-0.5
Circ. Q (SUS)		2.2	-2.2	-2.2	2.2	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		7.6	-7.6	-11.1	11.1	-2.4	2.4	-2.4	2.4
Circ. Q (TOTAL)		9.8	-9.8	-13.3	13.3	-2.4	2.4	-2.4	2.4

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Long. P1 (SUS)	1.6	1.6	-1.6	-1.6	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	5.3	5.3	-8.8	-8.8	-2.7	-2.7	-2.7	-2.7
Long. P1(TOTAL)	6.9	6.9	-10.4	-10.4	-2.7	-2.7	-2.7	-2.7
Long. Q (SUS)	7.8	-7.8	-7.8	7.8	0.0	0.0	-0.0	0.0
Long. Q (OCC)	25.8	-25.8	-42.2	42.2	-4.4	4.4	-4.4	4.4
Long. Q (TOTAL)	33.7	-33.7	-50.0	50.0	-4.4	4.4	-4.4	4.4
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-2.7	-2.7	2.7	2.7
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	23.7	24.1	23.7	24.1	23.7	24.1	23.7	24.1
Pm (SUS+OCC)	23.7	24.1	23.7	24.1	23.7	24.1	23.7	24.1
Pm+P1 (SUS)	27.8	28.2	19.6	20.0	24.1	24.4	24.1	24.4
Pm+P1 (SUS+OCC)	42.9	43.2	2.4	2.0	22.5	22.8	22.5	22.8
Pm+P1+Q (Total)	70.9	30.2	48.6	51.5	14.5	31.0	14.5	31.0

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.08	138.00	Passed
Pm (SUS+OCC)	24.08	165.60	Passed
Pm+P1 (SUS)	28.17	207.00	Passed
Pm+P1 (SUS+OCC)	43.24	248.40	Passed
Pm+P1+Q (TOTAL)	70.85	464.02	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 N./mm2
Baseplate Length	B 154.0000 mm.
Baseplate Width	D 154.0000 mm.
Baseplate Thickness	BTHK 13.0000 mm.
Leg Dimension Along Baseplate Length	d 76.2000 mm.
Leg Dimension Along Baseplate Width	b 76.2000 mm.
Bolt Circle Diameter	BCD 922.0000 mm.
Bolt Material	SA-36
Bolt Allowable Tensile Stress	STBA 114.00 N./mm2
Bolt Allowable Shear Stress	SBShear 99.93 N./mm2
Anchor Bolt Nominal Diameter	BOD 24.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB 0
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 1.5000 mm.
Ultimate 28-day Concrete Strength	FCPRIME 35.000 N./mm2

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 245/(2.26 • 4)
 = 2.7 N./mm². Must be less than 99.9 N./mm².

Plate thickness required (1/2"; 13mm) per D. Moss



Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle L3X3X0.2500

USA AISC 1989 Steel Table

Overall Leg Length		1475.000	mm.
Effective Leg Length	Leglen	1245.000	mm.
Distance Leg Up Side of Vessel		150.000	mm.
Number of Legs	Nleg	4	
Cross Sectional Area for L3X3X0.2500	Aleg	9.290	cm ²
Section Inertia (strong axis)		51.613	cm ⁴
Section Inertia (weak axis)		51.613	cm ⁴
Section Modulus (strong axis)		9455.335	mm. ³
Section Modulus (weak axis)		9455.335	mm. ³
Radius of Gyration (strong axis)		23.622	mm.
Radius of Gyration (weak axis)		23.622	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		47.4	Kg-m.
Total Weight Load at top of Legs	W	2158.4	Kgf
Total Shear force at top of Legs		80.8	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	150.000	mm.
Pad Length along Vessel Axis	C22P	150.000	mm.
Pad Thickness	Tpad	10.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (80.8) (82.6/207)$
 $= 32.19 \text{ Kgf}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (21166/4 + (465221 / (2 \cdot 520))) / 929$
 $= 6.18 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (2158/4) / 9.29$
 $= 5.70 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	76.200	76.200	6.3500
Distance to geometric centroid:	21.387	21.387	
Arm about YY:	18.212	19.888	
Arm about ZZ:	16.713	18.212	
Leg areas:	4.8387	4.4355	
Geometric inertia components YY:	16.211	35.578	
Geometric inertia components ZZ:	36.929	14.860	
Geometric inertias Iy & Iz:	51.789	51.789	
Product of inertia:	30.793		
Mohrs Radius:	30.793		
Average Inertia:	51.789		

QFACT =	1.0000	FBZ	=	0.14893
Principal Axis Inertias (Z&W)			=	20.996 82.582
Angle to Principal Axis			=	45.000



Distances to extreme fibers CW & CZ = 53.882 23.636
 FOB from Eq 5-5 = 0.99348
 Bending allowables Fby & Fbz = 0.16383 0.14893

Shear Center Coordinates Wo & Zo: 25.368 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.30000E-01 2.7256
 AREA, LENGTH, Kw, Kz: 1.4400 49.016 1.0000 1.0000
 H, Few, Fez, Fej: 0.63403 166.97 42.451 86.726
 Fe computed from C4-1: 68.965

Initial (Kl/r)max, & (Kl/r)equiv = 82.816 64.975
 Final (Kl/r)max, & Cc = 82.816 127.18
 Fa based on Eq 4-1 = 0.10424

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	31.29	198.53
Strong Axis Bending :	18.13	218.38
Axial Compression :	6.18	138.96

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.285

AISC Unity Check : 0.285, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Test Condition

Rectangular Attachment Parameter C11 107.763 mm.
 Rectangular Attachment Parameter C22 130.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	950.000	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	150.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	234.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	234.00	N./mm ²
Attachment Type	Type	Rectangular	
Parameter C11	C11	107.76	mm.
Parameter C22	C22	130.95	mm.
Thickness of Reinforcing Pad	Tpad	10.000	mm.
Pad Parameter C11P	C11p	150.000	mm.
Pad Parameter C22P	C22p	150.000	mm.
Design Internal Pressure	Dp	3.500	bars
Include Pressure Thrust		No	
Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	
Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	
Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	539.6	Kgf

Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	18.7	Kg-m.

Internal Pressure (SUS)	P	3.50	bars
Include Pressure Thrust		No	

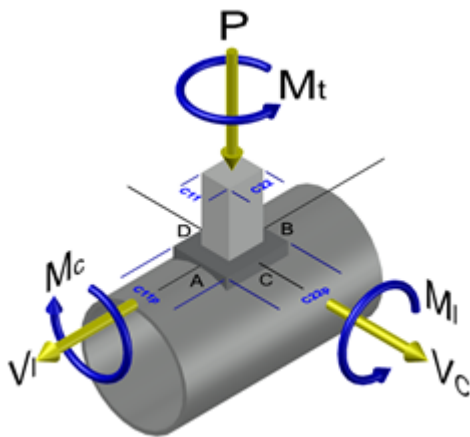
Global Force (OCC)	Fx	32.2	Kgf
Global Force (OCC)	Fy	45.7	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	21.6	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	8.85	bars
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	539.6	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-18.7	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 28.62

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.136	4C	4.819	(A,B)
N(PHI) / (P/Rm)	0.136	3C	3.938	(C,D)
M(PHI) / (P)	0.119	2C1	0.088	(A,B)
M(PHI) / (P)	0.119	1C	0.123	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.118	3A	0.873	(A,B,C,D)



M(PHI) / (MC/(Rm * Beta)) 0.131 1A 0.094 (A,B,C,D)
 N(PHI) / (ML/(Rm**2 * Beta)) 0.126 3B 2.990 (A,B,C,D)
 M(PHI) / (ML/(Rm * Beta)) 0.127 1B 0.043 (A,B,C,D)

 N(x) / (P/Rm) 0.128 3C 4.077 (A,B)
 N(x) / (P/Rm) 0.128 4C 4.906 (C,D)
 M(x) / (P) 0.129 1C1 0.118 (A,B)
 M(x) / (P) 0.129 2C 0.080 (C,D)
 N(x) / (MC/(Rm**2 * Beta)) 0.118 4A 1.280 (A,B,C,D)
 M(x) / (MC/(Rm * Beta)) 0.149 2A 0.048 (A,B,C,D)
 N(x) / (ML/(Rm**2 * Beta)) 0.126 4B 0.876 (A,B,C,D)
 M(x) / (ML/(Rm * Beta)) 0.135 2B 0.068 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.0	1.0	-1.0	-1.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.7	-2.7	-2.7	2.7	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		3.6	-1.7	-3.6	1.7	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		3.9	-3.9	-3.9	3.9	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		4.2	-3.6	-4.2	3.6	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Str. Int.		4.2	3.6	4.2	3.6	2.4	2.4	2.4	2.4

Dimensionless Parameters used : Gamma = 68.79

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.156	4C	9.591	(A,B)
N(PHI) / (P/Rm)	0.156	3C	6.193	(C,D)
M(PHI) / (P)	0.156	2C1	0.036	(A,B)
M(PHI) / (P)	0.156	1C !	0.070	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.156	3A	2.738	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.156	1A	0.074	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.156	3B	6.999	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.156	1B	0.025	(A,B,C,D)
N(x) / (P/Rm)	0.156	3C	6.193	(A,B)
N(x) / (P/Rm)	0.156	4C	9.591	(C,D)
M(x) / (P)	0.156	1C1	0.070	(A,B)
M(x) / (P)	0.156	2C !	0.038	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.156	4A	5.228	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.156	2A	0.036	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.156	4B	2.760	(A,B,C,D)



M(x) / (ML/(Rm * Beta)) 0.156 2B 0.032 (A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		5.1	5.1	-5.1	-5.1	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		7.5	-7.5	-7.5	7.5	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		12.5	-2.4	-12.5	2.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		2.0	2.0	-2.0	-2.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		9.7	-9.7	-9.7	9.7	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		11.7	-7.7	-11.7	7.7	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-2.5	-2.5	2.5	2.5
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-2.5	-2.5	2.5	2.5
Str. Int.		12.5	7.7	12.5	7.7	5.0	5.0	5.0	5.0

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	32.2	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	45.7	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-21.6	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 28.62

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Circ. Bend. P		-0.6	0.6	-0.6	0.6	-0.8	0.8	-0.8	0.8
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.1	1.1	-1.1	-1.1	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		3.1	-3.1	-3.1	3.1	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		3.4	-1.6	-5.0	2.4	-1.0	0.7	-1.0	0.7
Long. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Long. Bend. P		-0.8	0.8	-0.8	0.8	-0.5	0.5	-0.5	0.5
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.9	-0.9	-0.9	-0.9	-0.6	-0.6	-0.6	-0.6
Circ. Bend. P		-1.4	1.4	-1.4	1.4	-2.7	2.7	-2.7	2.7
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		5.9	5.9	-5.9	-5.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		8.6	-8.6	-8.6	8.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		12.2	-2.3	-16.8	3.3	-3.3	2.1	-3.3	2.1
Long. Memb. P		-0.6	-0.6	-0.6	-0.6	-0.9	-0.9	-0.9	-0.9
Long. Bend. P		-2.7	2.7	-2.7	2.7	-1.5	1.5	-1.5	1.5
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		2.3	2.3	-2.3	-2.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		11.2	-11.2	-11.2	11.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		10.3	-6.8	-16.8	11.0	-2.4	0.6	-2.4	0.6
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Str. Int.		12.2	6.8	16.8	11.0	3.3	2.2	3.3	2.2

Vessel Stress Summation at Attachment Junction (N./mm²)

[illegible]



Long. Pm(TOTAL)	17.1	17.1	17.1	17.1	17.1	17.1	17.1	17.1
Long. P1 (SUS)	0.3	0.3	-0.3	-0.3	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.2	0.2	-0.5	-0.5	-0.2	-0.2	-0.2	-0.2
Long. P1(TOTAL)	0.5	0.5	-0.8	-0.8	-0.2	-0.2	-0.2	-0.2
Long. Q (SUS)	3.9	-3.9	-3.9	3.9	0.0	0.0	-0.0	0.0
Long. Q (OCC)	3.8	-3.8	-5.3	5.3	-0.5	0.5	-0.5	0.5
Long. Q (TOTAL)	7.7	-7.7	-9.2	9.2	-0.5	0.5	-0.5	0.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	9.7	10.0	9.7	10.0	9.7	10.0	9.7	10.0
Pm (SUS+OCC)	34.1	35.4	34.1	35.4	34.1	35.4	34.1	35.4
Pm+P1 (SUS)	10.6	11.0	8.7	9.1	9.9	10.3	9.9	10.3
Pm+P1 (SUS+OCC)	36.0	37.2	31.9	33.1	34.1	35.3	34.1	35.3
Pm+P1+Q (Total)	41.2	32.1	25.5	39.4	33.3	36.1	33.3	36.1

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.02	234.00	Passed
Pm (SUS+OCC)	35.36	280.79	Passed
Pm+P1 (SUS)	10.98	350.99	Passed
Pm+P1 (SUS+OCC)	37.24	421.19	Passed
Pm+P1+Q (TOTAL)	41.18	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		23.7	24.1	23.7	24.1	23.7	24.1	23.7	24.1
Circ. Pm (OCC)		60.0	60.9	60.0	60.9	60.0	60.9	60.0	60.9
Circ. Pm(TOTAL)		83.7	85.0	83.7	85.0	83.7	85.0	83.7	85.0
Circ. P1 (SUS)		5.1	5.1	-5.1	-5.1	0.0	0.0	-0.0	-0.0
Circ. P1 (OCC)		5.0	5.0	-6.8	-6.8	-0.6	-0.6	-0.6	-0.6
Circ. P1(TOTAL)		10.0	10.0	-11.8	-11.8	-0.6	-0.6	-0.6	-0.6
Circ. Q (SUS)		7.5	-7.5	-7.5	7.5	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		7.2	-7.2	-10.0	10.0	-2.7	2.7	-2.7	2.7
Circ. Q (TOTAL)		14.7	-14.7	-17.5	17.5	-2.7	2.7	-2.7	2.7
Long. Pm (SUS)		11.9	11.9	11.9	11.9	11.9	11.9	11.9	11.9
Long. Pm (OCC)		30.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0
Long. Pm(TOTAL)		41.9	41.9	41.9	41.9	41.9	41.9	41.9	41.9
Long. P1 (SUS)		2.0	2.0	-2.0	-2.0	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)		1.7	1.7	-2.9	-2.9	-0.9	-0.9	-0.9	-0.9
Long. P1(TOTAL)		3.7	3.7	-4.9	-4.9	-0.9	-0.9	-0.9	-0.9



Long. Q (SUS)	9.7	-9.7	-9.7	9.7	0.0	0.0	-0.0	0.0
Long. Q (OCC)	8.5	-8.5	-13.9	13.9	-1.5	1.5	-1.5	1.5
Long. Q (TOTAL)	18.2	-18.2	-23.6	23.6	-1.5	1.5	-1.5	1.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-2.5	-2.5	2.5	2.5
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-2.7	-2.7	2.7	2.7
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	23.7	24.1	23.7	24.1	23.7	24.1	23.7	24.1
Pm (SUS+OCC)	83.7	85.0	83.7	85.0	83.7	85.0	83.7	85.0
Pm+P1 (SUS)	28.8	29.1	18.7	19.0	24.2	24.6	24.2	24.6
Pm+P1 (SUS+OCC)	93.8	95.0	71.9	73.1	83.3	84.6	83.3	84.6
Pm+P1+Q (Total)	108.5	80.3	54.4	90.6	80.6	87.3	80.6	87.3

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.08	234.00	Passed
Pm (SUS+OCC)	84.96	280.79	Passed
Pm+P1 (SUS)	29.15	350.99	Passed
Pm+P1 (SUS+OCC)	95.00	421.19	Passed
Pm+P1+Q (TOTAL)	108.46	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material		SA-516 70
Baseplate Allowable Stress	SBA	137.90 N./mm2
Baseplate Length	B	154.0000 mm.
Baseplate Width	D	154.0000 mm.
Baseplate Thickness	BTHK	13.0000 mm.
Leg Dimension Along Baseplate Length	d	76.2000 mm.
Leg Dimension Along Baseplate Width	b	76.2000 mm.
Bolt Circle Diameter	BCD	922.0000 mm.
Bolt Material		SA-36
Bolt Allowable Tensile Stress	STBA	114.00 N./mm2
Bolt Allowable Shear Stress	SBShear	99.93 N./mm2
Anchor Bolt Nominal Diameter	BOD	24.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB	0
Total Number of Anchors Bolt per Leg	NBT	1
Bolt Corrosion Allowance	bca	1.5000 mm.
Ultimate 28-day Concrete Strength	FCPRIME	35.000 N./mm2

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 245/(2.26 • 4)
 = 2.7 N./mm². Must be less than 99.9 N./mm².

Plate thickness required (1/2"; 13mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be



selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: B23"

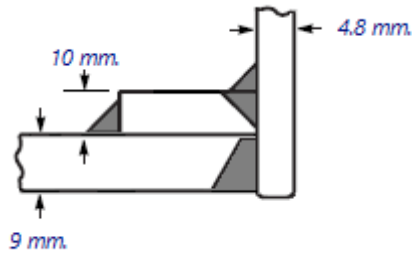
From: 10

Pressure for Reinforcement Calculations	P	3.623	bars
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	950.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	9.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	131.6740	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	9.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B23"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.189 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.62 \cdot 956 \cdot 0.89 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.62)$
 $= 1.1230$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.62 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 3.62)$
 $= 0.1363$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3358 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	170.5976 mm.
Parallel to Vessel Wall, opening length	d	85.2988 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		14.5015 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: B23".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3358 mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 4.2449 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000 mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.3553 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm.
Wall Thickness per table UG-45	tb3	= 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.34, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 4.8006 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained	:	106.7,	Allowable	:	118.0 N./mm2	Passed
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Expansion :	0.0,	Allowable :	188.3 N./mm2	Passed
Occasional :	4.2,	Allowable :	156.9 N./mm2	Passed
Shear :	77.3,	Allowable :	82.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 4.8, tr = 0.14, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 4.8, tr = 0.14, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 4.8, tr = 0.14, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9, tr = 1.12, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 4.8, tr = 0.14, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.62/51.00 = 0.071$

Weld Size Calculations, Description: B23"

Intermediate Calc. for nozzle/shell Welds	Tmin	1.8006 mm.
Intermediate Calc. for pad/shell Welds	TminPad	9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.2604 = 0.7 * tmin.$	$6.3630 = 0.7 * Wo mm.$



Pad Weld 4.5000 = 0.5*TminPad 5.6560 = 0.7 * Wp mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.1465 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 141.8206 mm.

Input Echo, WRC107/537 Item 1, Description: B23" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	1710.000	mm.
Vessel Thickness	Tv	9.000	mm.
Design Temperature	T1	150.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	88.900	mm.
Nozzle Thickness	Tn	4.801	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2
Thickness of Reinforcing Pad	Tpad	10.000	mm.
Diameter of Reinforcing Pad	Dpad	152.400	mm.
Design Internal Pressure	Dp	3.623	bars
Include Pressure Thrust		No	

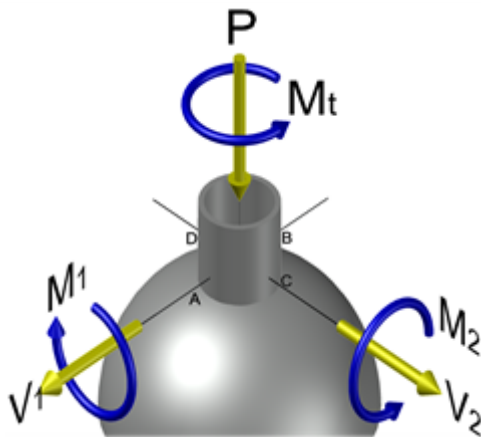
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-489.5	Kgf
Longitudinal Shear (SUS)	(V1) V1	432.7	Kgf
Circumferential Shear (SUS)	(Vc) V2	432.7	Kgf
Circumferential Moment (SUS)	(Mc) M1	85.1	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	85.1	Kg-m.
Torsional Moment (SUS)	Mt	110.1	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(861.0 (9.0 - 3.0))}$
 $= 326.087 \text{ mm.}$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-489.5	Kgf
Circumferential Shear	(VC) V2	432.7	Kgf
Longitudinal Shear	(VL) V1	432.7	Kgf
Circumferential Moment	(MC) M1	85.1	Kg-m.
Longitudinal Moment	(ML) M2	85.1	Kg-m.
Torsional Moment	MT	110.1	Kg-m.

Dimensionless Parameters: U = 0.38 TAU = 24.19 RHO = 8.89

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 7	0.02549	(A,B,C,D)
$M(x) / P$	SP 7	0.00805	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.09244	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 7	0.02547	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.09244	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 7	0.02547	(A,B,C,D)
$N(y) * T / P$	SP 7	0.37861	(A,B,C,D)
$M(y) / P$	SP 7	0.11558	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.44312	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 7	0.44170	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.44312	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 7	0.44170	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Rad. Bend.	P	0.9	-0.9	0.9	-0.9	0.9	-0.9	0.9	-0.9
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-2.6	-2.6	2.6	2.6
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-4.2	4.2	4.2	-4.2
Rad. Memb.	ML	-2.6	-2.6	2.6	2.6	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-4.2	4.2	4.2	-4.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-5.4	1.2	8.2	-2.1	-5.4	1.2	8.2	-2.1



Tang. Memb. P	7.1	7.1	7.1	7.1	7.1	7.1	7.1	7.1
Tang. Bend. P	13.0	-13.0	13.0	-13.0	13.0	-13.0	13.0	-13.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-12.3	-12.3	12.3	12.3
Tang. Bend. MC	0.0	0.0	0.0	0.0	-73.4	73.4	73.4	-73.4
Tang. Memb. ML	-12.3	-12.3	12.3	12.3	0.0	0.0	0.0	0.0
Tang. Bend. ML	-73.4	73.4	73.4	-73.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-65.6	55.2	105.8	-67.0	-65.6	55.2	105.8	-67.0
Shear VC	1.9	1.9	-1.9	-1.9	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.9	-1.9	1.9	1.9
Shear MT	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4
Tot. Shear	7.3	7.3	3.5	3.5	3.5	3.5	7.3	7.3
Str. Int.	66.5	56.2	105.9	67.2	65.8	55.5	106.3	67.9

Unitless Prm: U = 1.06 TAU = 0.00 (41.82) RHO = 0.00 (3.33)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.06096	(A,B,C,D)
M(x) / P	SR 2	0.03175	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.07029	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.06364	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.07029	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.06364	(A,B,C,D)
N(y) * T / P	SR 2	0.01850	(A,B,C,D)
M(y) / P	SR 2	0.00976	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.02085	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.01919	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.02085	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.01919	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		8.1	8.1	8.1	8.1	8.1	8.1	8.1	8.1
Rad. Bend. P		25.4	-25.4	25.4	-25.4	25.4	-25.4	25.4	-25.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-22.7	-22.7	22.7	22.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-123.2	123.2	123.2	-123.2
Rad. Memb. ML		-22.7	-22.7	22.7	22.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-123.2	123.2	123.2	-123.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-112.3	83.2	179.4	-117.8	-112.3	83.2	179.4	-117.8
Tang. Memb. P		2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Tang. Bend. P		7.8	-7.8	7.8	-7.8	7.8	-7.8	7.8	-7.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-6.7	-6.7	6.7	6.7
Tang. Bend. MC		0.0	0.0	0.0	0.0	-37.1	37.1	37.1	-37.1
Tang. Memb. ML		-6.7	-6.7	6.7	6.7	0.0	0.0	0.0	0.0
Tang. Bend. ML		-37.1	37.1	37.1	-37.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-33.6	25.1	54.1	-35.8	-33.6	25.1	54.1	-35.8
Shear VC		3.0	3.0	-3.0	-3.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-3.0	-3.0	3.0	3.0
Shear MT		4.9	4.9	4.9	4.9	4.9	4.9	4.9	4.9
Tot. Shear		7.9	7.9	2.0	2.0	2.0	2.0	7.9	7.9



Str. Int.	113.1	84.3	179.4	117.8	112.4	83.3	179.9	118.5
-----------	-------	------	-------	-------	-------	------	-------	-------

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7
Rad.	Pl (SUS)	-2.1	-2.1	3.0	3.0	-2.1	-2.1	3.0	3.0
Rad.	Q (SUS)	-3.3	3.3	5.1	-5.1	-3.3	3.3	5.1	-5.1
Long.	Pm (SUS)	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7
Long.	Pl (SUS)	-5.2	-5.2	19.4	19.4	-5.2	-5.2	19.4	19.4
Long.	Q (SUS)	-60.4	60.4	86.4	-86.4	-60.4	60.4	86.4	-86.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.9	1.9	-1.9	-1.9	-1.9	-1.9	1.9	1.9
Shear	Q (SUS)	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4
	Pm (SUS)	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7
	Pm+Pl (SUS)	8.5	8.5	29.3	29.3	8.5	8.5	29.3	29.3
	Pm+Pl+Q (Total)	61.9	65.9	115.6	65.3	60.6	65.2	116.1	66.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.72	138.00	Passed
Pm+Pl (SUS)	29.31	207.00	Passed
Pm+Pl+Q (TOTAL)	116.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9
Rad.	Pl (SUS)	-14.5	-14.5	30.8	30.8	-14.5	-14.5	30.8	30.8
Rad.	Q (SUS)	-97.8	97.8	148.6	-148.6	-97.8	97.8	148.6	-148.6
Long.	Pm (SUS)	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9
Long.	Pl (SUS)	-4.3	-4.3	9.2	9.2	-4.3	-4.3	9.2	9.2
Long.	Q (SUS)	-29.3	29.3	45.0	-45.0	-29.3	29.3	45.0	-45.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	3.0	3.0	-3.0	-3.0	-3.0	-3.0	3.0	3.0
Shear	Q (SUS)	4.9	4.9	4.9	4.9	4.9	4.9	4.9	4.9
	Pm (SUS)	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9
	Pm+Pl (SUS)	22.4	22.4	57.1	57.1	22.4	22.4	57.1	57.1
	Pm+Pl+Q (Total)	87.2	110.2	205.3	91.9	86.5	109.2	205.8	92.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.90	138.00	Passed
Pm+Pl (SUS)	57.11	207.00	Passed
Pm+Pl+Q (TOTAL)	205.79	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: L22"

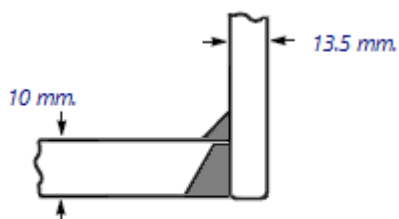
From: 20

Pressure for Reinforcement Calculations	P	3.583	bars
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	950.00	mm.
Design Length of Section	L	1733.3334	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.58 \cdot 478 / (138 \cdot 1 - 0.6 \cdot 3.58)$
 $= 1.2430$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.58 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 3.58)$
 $= 0.0738$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3775 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3775	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 4.2430	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.3584	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.38, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 13.4620 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10, tr = 1.24, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 3.58/19.60 = 0.183

Weld Size Calculations, Description: L22"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.5924 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.1425 mm.

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Input, Nozzle Desc: L12"

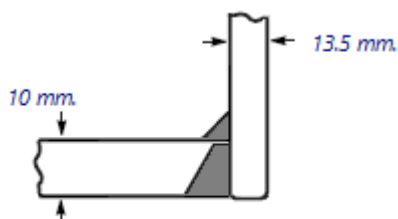
From: 20

Pressure for Reinforcement Calculations	P	3.514	bars
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	950.00	mm.
Design Length of Section	L	1733.3334	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1050.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 478 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 1.2190$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 0.0724$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3775 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3775	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 4.2190	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.3584	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.38, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 13.4620 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10, tr = 1.22, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.17$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 3.51/19.60 = 0.179

Weld Size Calculations, Description: L12"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.9000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.5924 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.1425 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm.

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Input, Nozzle Desc: A24"

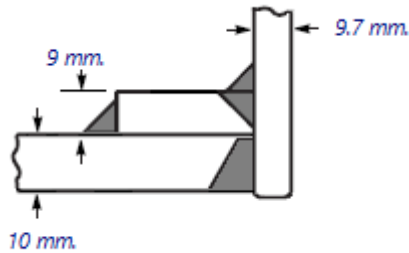
From: 20

Pressure for Reinforcement Calculations	P	3.532	bars
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	950.00	mm.
Design Length of Section	L	1733.3334	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	9.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	203.2000	mm.
Thickness of Pad	te	9.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	9.0000	mm.
Reinforcing Pad Width		44.4500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.53 \cdot 478 / (138 \cdot 1 - 0.6 \cdot 3.53)$
 $= 1.2254$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.53 \cdot 57.1 / (118 \cdot 1 + 0.4 \cdot 3.53)$
 $= 0.1709$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4718 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	201.6618 mm.
Parallel to Vessel Wall, opening length	d	100.8309 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		17.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

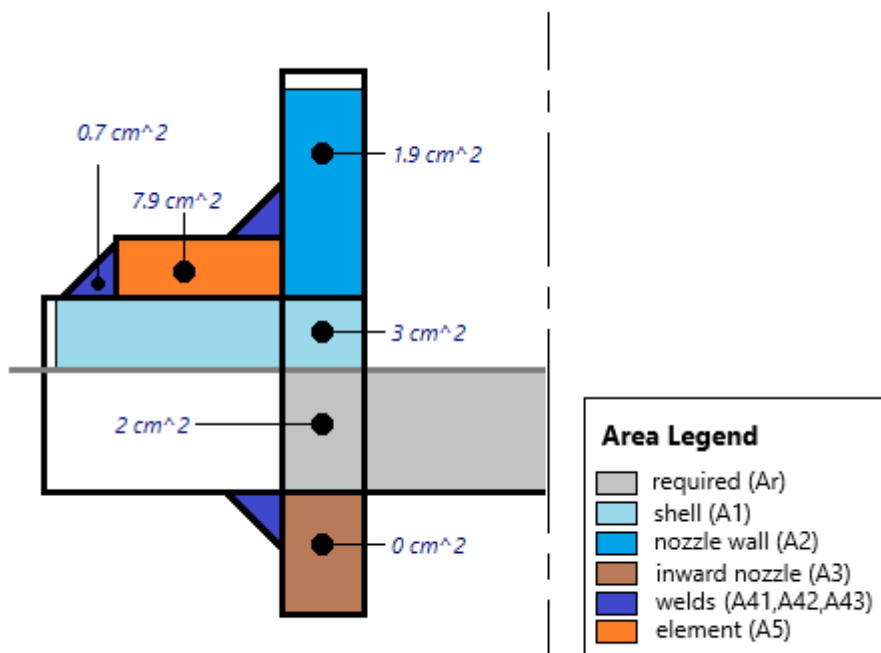


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1.260	2.037	NA
Area in Shell	A1	5.710	3.002	NA
Area in Nozzle Wall	A2	1.964	1.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.690	0.690	NA
Area in Element	A5	7.863	7.863	NA
TOTAL AREA AVAILABLE	Atot	16.227	13.430	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1)) \text{ per UG-37(d)}$$

$$= 0.5(101 \cdot 3.96 \cdot 1 + 2 \cdot 6.73 \cdot 3.96 \cdot 1(1-0.86))$$

$$= 2.037 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 101(1 \cdot 7 - 1 \cdot 3.96) - 2 \cdot 6.73$$

$$(1 \cdot 7 - 1 \cdot 3.96) \cdot (1 - 0.86)$$

$$= 3.002 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 17.5) \cdot (6.73 - 0.47) \cdot 0.86$$

$$= 1.874 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$$

$$= (0.81) \cdot 0.86 + (0) \cdot 0.86 + 0^2 \cdot 1$$



= 0.690 cm²

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_r$$

$$= (202 - 114) \cdot 9 \cdot 1$$

$$= 7.863 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4718 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $tr_{b1} = 4.2254 \text{ mm.}$
 Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $tr_{b2} = 3.3584 \text{ mm.}$
 Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.26, \max(4.5, 4.5)]$$

$$= 4.5000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.47, 4.5)$$

$$= 4.5000 \text{ mm.}$$

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	34.7,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	260.3 N./mm ²	Passed
Occasional	:	1.2,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	25.3,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.73$, $tr = 0.17$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 9, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.18$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 0.17, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.53/51.00 = 0.069$

Weld Size Calculations, Description: A24"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm.
 Intermediate Calc. for pad/shell Welds TminPad 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 * t_{min}$	$6.3630 = 0.7 * W_o$ mm.
Pad Weld	$3.5000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2.04 - 3 + 2 \cdot 6.73 \cdot 0.86 \cdot (1 \cdot 7 - 3.96)) \cdot 138)$
 $= \max(0, -866.27) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (1.87 + 7.86 + 0.69 - 0 \cdot 0.86) \cdot 138$
 $= 14672.89 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (1.87 + 0 + 0.69 + (0.81)) \cdot 138$
 $= 4746.48 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (1.87 + 0 + 0.69 + 7.86 + (0.81)) \cdot 138$
 $= 15807.33 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 114 \cdot 9 \cdot 0.49 \cdot 118$
 $= 9527. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:



$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 203 \cdot 8 \cdot 0.49 \cdot 138$$

$$= 17606. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 53.8) \cdot (9.73 - 3) \cdot 0.7 \cdot 118$$

$$= 9584. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 114 \cdot 9 \cdot 0.74 \cdot 138$$

$$= 16826. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 114 \cdot (10 - 3) \cdot 0.74 \cdot 138$$

$$= 13087. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (17606 + 9584) = 27190 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9527 + 16826 + 13087 + 0) = 39440 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (17606 + 13087 + 0) = 30693 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 27190	Kgf, must exceed W = 0 Kg	or W1 = 14672 Kg
Path 2-2 = 39439	Kgf, must exceed W = 0 Kg	or W2 = 4746 Kg
Path 3-3 = 30693	Kgf, must exceed W = 0 Kg	or W3 = 15807 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 16.9 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 3.4506 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.0006 mm.

Input Echo, WRC297 Item 1, Description: A24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	950.000	mm.
Shell Thickness	10.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	114.300	mm.
Nozzle Thickness	9.7345	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	



Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	203.200	mm.
Thickness of Reinforcing Pad	9.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	3.53	bars
Radial Load	P	-652.66	Kgf
Circumferential Shear	Vc	489.50	Kgf
Longitudinal Shear	Vl	652.66	Kgf
Circumferential Moment	Mc	130.48	Kg-m.
Longitudinal Moment	Ml	169.63	Kg-m.
Torsional Moment	Mt	195.72	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(486.0(10.0 - 3.0))}$
 $= 306.778 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	972.000	mm.
Nozzle Outside Diameter	(d) =	114.300	mm.
Vessel Thickness used	(T) =	16.000	mm.
Nozzle Thickness used	(t) =	6.735	mm.
T / t	=	2.376	
d / t	=	16.972	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.917$$

Nr/P	=	0.120
Mr/P	=	0.127
M0/P	=	0.093
N0/P	=	0.182
MrD/Mc	=	0.203
NrDL/Mc	=	0.159
M0d/Mc	=	0.171
N0DL/Mc	=	0.080
MrD/Ml	=	0.164
NrDL/Ml	=	0.134
M0D/Ml	=	0.136
N0DL/Ml	=	0.187

Vessel Stresses (N./mm2):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-10	-10	10	10
Outplane Bending (ML)	-46	46	46	-46
Normal Pressure Stress	10	10	10	10



Outplane Stress Summary -28 37 86 -34

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	5	5	5	5
Inplane Stress Summary	-36	37	90	-59

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	40	44	93	59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	5	5	5	5
Outplane Stress Summary	-24	37	72	-45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-53	53	53	-53
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	-27	40	92	-51

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	8	8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	30	43	95	57

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-10	-10	10	10
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3
Hoop Stress Summary	-3	-2	18	18

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	56	-56	56	-56
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-28	-28	28	28
Axial Bending (ML)	-186	186	186	-186
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-154	105	275	-209

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13
Shear Stress Summary	17	17	9	9

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	156	113	276	229

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	2	3	2	3
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Hoop Stress Summary	4	4	11	11
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	2	2	2	2
Axial Bending (P)	56	-56	56	-56
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-181	181	181	-181
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-143	107	264	-211
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	13	13	13	13

Shear Stress Summary	8	8	19	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	148	108	266	226
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{481.5(10.0 - 3.0)}$$

$$= 305.885 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 963.000 mm.

Nozzle Outside Diameter (d) = 203.200 mm.

Vessel Thickness used (T) = 7.000 mm.

Nozzle Thickness used (t) = 51.185 mm.

T / t = 0.137

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.475$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.069

Mr/P = 0.125

M0/P = 0.037

N0/P = 0.055

MrD/Mc = 0.280

NrDL/Mc = 0.105

M0d/Mc = 0.085

N0DL/Mc = 0.071

MrD/Ml = 0.095

NrDL/Ml = 0.092

M0D/Ml = 0.028

N0DL/Ml = 0.073

Vessel Stresses (N./mm²):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	23	24	23	24
Outplane Stress Summary	19	18	100	-13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	97	-97	97	-97
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-15	-15	15	15
Inplane Bending (ML)	-94	94	94	-94
Inplane Pressure Stress	11	11	11	11
Inplane Stress Summary	8	2	228	-156

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	22	20	228	156

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-65	65	65	-65
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11	11	11	11
Outplane Stress Summary	-26	46	122	-65

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	97	-97	97	-97
Inplane Membrane (Mc)	-13	-13	13	13



Inplane Bending (Mc)	-215	215	215	-215
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	23	24	23	24

Inplane Stress Summary	-98	138	359	-266
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2	2
Torsional Shear (Mt)	4	4	4	4	4
Shear Stress Summary	1	1	7	7	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	98	138	360	266	266

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	10	10	10	10	10	10	10
Circ. Pl (SUS)	-6	-6	14	14	-4	-4	8	8
Circ. Q (SUS)	-32	32	60	-60	-35	35	71	-71
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-5	-5	9	9	1	1	7	7
Long. Q (SUS)	-37	37	73	-73	-30	30	58	-58
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
Pm+Pl (SUS)	4.5	4.5	24.1	24.1	8.0	8.0	18.6	18.6
Pm+Pl+Q (Total)	40.0	42.5	89.8	59.7	30.4	42.4	91.3	57.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	138.00	Passed
Pm+Pl (SUS)	24.10	207.00	Passed
Pm+Pl+Q (TOTAL)	91.30	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	23	24	23	24	23	24	23	24
Circ. Pl (SUS)	-5	-5	19	19	-4	-4	22	22
Circ. Q (SUS)	0	0	56	-56	-118	118	312	-312
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-6	-6	24	24	-2	-2	16	16
Long. Q (SUS)	3	-3	191	-191	-37	37	93	-93
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	23.0	24.0	23.0	24.0	23.0	24.0	23.0	24.0
Pm+Pl (SUS)	18.3	19.3	42.5	43.5	19.4	20.4	45.2	46.2
Pm+Pl+Q (Total)	20.8	20.9	226.0	156.0	99.1	138.0	357.2	266.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	138.00	Passed
Pm+Pl (SUS)	46.21	207.00	Passed
Pm+Pl+Q (TOTAL)	357.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	-6	-6	14	14	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-26	-26	30	30	-20	-20	24	24
Long. Q (SUS)	-130	130	242	-242	-125	125	237	-237
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-5	-5	5	5
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	25.4	25.4	31.6	31.6	24.2	25.1	26.4	26.5
Pm+Pl+Q (Total)	156.7	112.6	273.4	228.9	147.9	106.6	263.3	224.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	118.00	Passed
Pm+Pl (SUS)	31.62	177.00	Passed
Pm+Pl+Q (TOTAL)	273.39	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: A23"

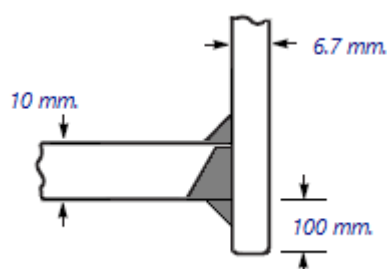
From: 20

Pressure for Reinforcement Calculations	P	3.532	bars
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	950.00	mm.
Design Length of Section	L	1733.3334	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	9.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	100.0000	mm.
Weld leg size, Inside Element to Shell	Wi	8.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A23"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 3.500 in.
 Actual Thickness Used in Calculation 0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.53 \cdot 478 / (138 \cdot 1 - 0.6 \cdot 3.53)$
 $= 1.2254$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.53 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 3.53)$
 $= 0.1329$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5037 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	163.1300	mm.
Parallel to Vessel Wall, opening length	d	81.5650	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.1688	mm.
Normal to Vessel Wall, Inward		1.6688	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: A23".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5037	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 4.2254	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.3584	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.5, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 6.67, tr = 0.13, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.036$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 6.67, tr = 0.13, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.036$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.53/102.00 = 0.035$

Weld Size Calculations, Description: A23"

Intermediate Calc. for nozzle/shell Welds	Tmin	3.6675	mm.
Intermediate Calc. for Inward Weld	TminIns	3.6675	mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.5673 = 0.7 * t_{min}$	$6.3630 = 0.7 * W_o$ mm.
Inward Weld	$2.5673 = 0.7 * T_{minIn}$	$2.6560 = 0.7 * W_{i-Can}$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.0844 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 460.0000 mm.

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Input, Nozzle Desc: A12"

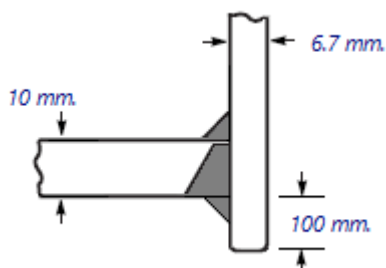
From: 20

Pressure for Reinforcement Calculations	P	3.532	bars
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	950.00	mm.
Design Length of Section	L	1733.3334	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	131.6740	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	100.0000	mm.
Weld leg size, Inside Element to Shell	Wi	8.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 3.500 in.
 Actual Thickness Used in Calculation 0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.53 \cdot 478 / (138 \cdot 1 - 0.6 \cdot 3.53)$
 $= 1.2254$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.53 \cdot 44.5 / (138 \cdot 1 + 0.4 \cdot 3.53)$
 $= 0.1137$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3358 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	163.1300	mm.
Parallel to Vessel Wall, opening length	d	81.5650	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.1688	mm.
Normal to Vessel Wall, Inward		1.6688	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: A12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3358	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 4.2254	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.3584	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.34, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	24.7,	Allowable	:	138.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	320.3 N./mm2	Passed
Occasional	:	1.9,	Allowable	:	183.5 N./mm2	Passed
Shear	:	21.0,	Allowable	:	96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 6.67, tr = 0.11, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.031$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 6.67, tr = 0.11, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.031, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.53/19.60 = 0.180

Weld Size Calculations, Description: A12"

Intermediate Calc. for nozzle/shell Welds Tmin 3.6675 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	2.5673 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.0844 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.6740 mm.

Input Echo, WRC297 Item 1, Description: A12" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	950.000	mm.
Shell Thickness	10.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	88.900	mm.
Nozzle Thickness	6.6675	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.53	bars
Radial Load	P	-326.33	Kgf



Circumferential Shear	Vc	244.75	Kgf
Longitudinal Shear	Vl	326.33	Kgf
Circumferential Moment	Mc	32.62	Kg-m.
Longitudinal Moment	Ml	42.41	Kg-m.
Torsional Moment	Mt	48.93	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(481.5(10.0 - 3.0))}$
 $= 280.485 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	963.000 mm.
Nozzle Outside Diameter	(d) =	88.900 mm.
Vessel Thickness used	(T) =	7.000 mm.
Nozzle Thickness used	(t) =	3.668 mm.
T / t	=	1.909
d / t	=	24.240

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.083$$

Nr/P	=	0.094
Mr/P	=	0.130
M0/P	=	0.075
N0/P	=	0.167
MrD/Mc	=	0.219
NrDL/Mc	=	0.135
M0d/Mc	=	0.145
N0DL/Mc	=	0.108
MrD/Ml	=	0.166
NrDL/Ml	=	0.109
M0D/Ml	=	0.102
N0DL/Ml	=	0.224

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-21	-21	21	21
Outplane Bending (ML)	-58	58	58	-58
Normal Pressure Stress	23	24	23	24
Outplane Stress Summary	-15	42	143	-31

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	50	-50	50	-50
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-94	94	94	-94
Inplane Pressure Stress	11	11	11	11



Inplane Stress Summary -36 51 174 -117

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		2	2	-2	-2
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		5	5	5	5
Shear Stress Summary		7	7	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		39	56	174	117

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		10	10	10	10
Outplane Bending (P)		29	-29	29	-29
Outplane Membrane (Mc)		-7	-7	7	7
Outplane Bending (Mc)		-63	63	63	-63
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		11	11	11	11
Outplane Stress Summary		-19	49	124	-62

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		6	6	6	6
Inplane Bending (P)		50	-50	50	-50
Inplane Membrane (Mc)		-9	-9	9	9
Inplane Bending (Mc)		-96	96	96	-96
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		23	24	23	24
Inplane Stress Summary		-25	66	187	-106

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-3	-3	3	3
Torsional Shear (Mt)		5	5	5	5
Shear Stress Summary		2	2	8	8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		26	66	188	108

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
--------------------	--	----	----	----	----



(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	4	4	4
Hoop Stress Summary	-5	-5	36	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	118	-118	118	-118
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-232	232	232	-232
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-129	98	376	-324

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	15	15	5	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	130	108	376	361

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	4	4	4
Hoop Stress Summary	7	7	23	23

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	118	-118	118	-118
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-242	242	242	-242
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress 2 2 2 2

Axial Stress Summary -134 114 382 -339

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	10	10	10	10
Shear Stress Summary	4	4	16	16

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	142	114	383	365

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	23	24	23	24	23	24	23	24
Circ. Pl (SUS)	-11	-11	31	31	-3	-3	15	15
Circ. Q (SUS)	-29	29	87	-87	-46	46	146	-146
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-4	-4	16	16	3	3	17	17
Long. Q (SUS)	-44	44	144	-144	-34	34	92	-92
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	23.0	24.0	23.0	24.0	23.0	24.0	23.0	24.0
Pm+Pl (SUS)	12.7	13.6	54.1	55.1	21.2	22.1	38.8	39.8
Pm+Pl+Q (Total)	39.2	54.8	171.3	117.1	26.6	67.2	185.0	108.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	138.00	Passed
Pm+Pl (SUS)	55.14	207.00	Passed
Pm+Pl+Q (TOTAL)	184.98	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	4	4	4	4	4	4	4
Circ. Pl (SUS)	-11	-11	31	31	3	3	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-17	-17	23	23	-12	-12	18	18
Long. Q (SUS)	-114	114	350	-350	-124	124	360	-360
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-6	-6	6	6
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pm+Pl (SUS)	16.7	16.7	36.4	36.4	20.8	20.8	26.5	26.5
Pm+Pl+Q (Total)	130.6	109.6	375.1	360.2	141.2	114.1	380.7	362.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	138.00	Passed
Pm+Pl (SUS)	36.40	207.00	Passed
Pm+Pl+Q (TOTAL)	380.71	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: B18"

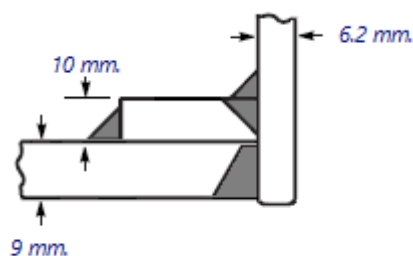
From: 30

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	950.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	9.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	112.6240	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	250.0000	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		40.8625	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.245 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 956 \cdot 0.89 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.5)$
 $= 1.0849$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 84.1 / (118 \cdot 1 + 0.4 \cdot 3.5)$
 $= 0.2493$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4557 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	323.6580 mm.
Parallel to Vessel Wall, opening length	d	161.8290 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

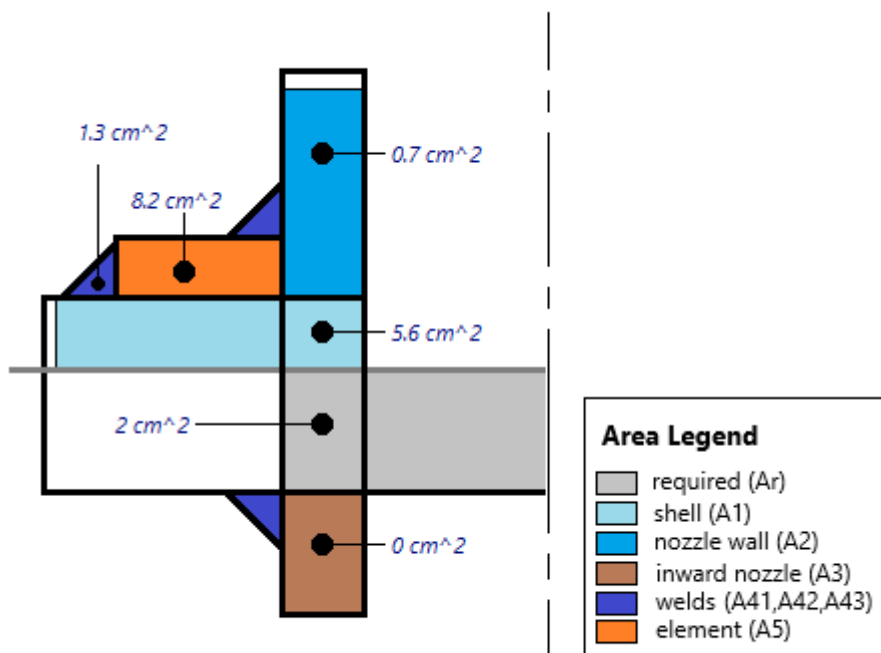


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1.766	2.040	NA
Area in Shell	A1	7.908	5.620	NA
Area in Nozzle Wall	A2	0.763	0.710	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.281	1.281	NA
Area in Element	A5	8.173	8.173	NA
TOTAL AREA AVAILABLE	Atot	18.125	15.783	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1)) \text{ per UG-37(d)} \\
 &= 0.5(162 \cdot 2.51 \cdot 1 + 2 \cdot 3.22 \cdot 2.51 \cdot 1(1-0.86)) \\
 &= 2.040 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 162(1 \cdot 6 - 1 \cdot 2.51) - 2 \cdot 3.22 \\
 &\quad (1 \cdot 6 - 1 \cdot 2.51) \cdot (1 - 0.86) \\
 &= 5.620 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 15) \cdot (3.22 - 0.46) \cdot 0.86 \\
 &= 0.710 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi-can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4 \\
 &= (0.75) \cdot 0.86 + (0) \cdot 0.86 + 8^2 \cdot 1
 \end{aligned}$$



= 1.281 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
 = (250 - 168) • 10 • 1
 = 8.173 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.4557 mm.
 Wall Thickness per UG16(b), tr16b = 4.5000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 4.2027 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 4.5000 mm.
 Wall Thickness, shell/head, external pressure trb2 = 3.3553 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm.
 Wall Thickness per table UG-45 tb3 = 9.2200 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[9.22, max(4.5, 4.5)]
 = 4.5000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.46, 4.5)
 = 4.5000 mm.

Available Nozzle Neck Thickness = 6.2230 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	120.7,	Allowable	:	118.0 N./mm2	Failed
Expansion	:	0.0,	Allowable	:	177.0 N./mm2	Passed
Occasional	:	4.3,	Allowable	:	156.9 N./mm2	Passed
Shear	:	77.4,	Allowable	:	82.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 6.22, tr = 0.25, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.077, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 6.22, tr = 0.25, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.077, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 6.22, tr = 0.25, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.077, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9, tr = 1.08, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.18, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 6.22, tr = 0.25, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.077, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/102.00 = 0.034

Weld Size Calculations, Description: B18"

Intermediate Calc. for nozzle/shell Welds Tmin 3.2230 mm.
 Intermediate Calc. for pad/shell Welds TminPad 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	2.2561 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	4.5000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2.04 - 5.62 + 2 \cdot 3.22 \cdot 0.86 \cdot (1 \cdot 6 - 2.51)) \cdot 138)$
 $= \max(0, -4765.70) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (0.71 + 8.17 + 1.28 - 0 \cdot 0.86) \cdot 138$
 $= 14301.91 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (0.71 + 0 + 0.82 + (0.33)) \cdot 138$
 $= 2622.10 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (0.71 + 0 + 1.28 + 8.17 + (0.33)) \cdot 138$
 $= 14767.26 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$
 $= 15584. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 250 \cdot 8 \cdot 0.49 \cdot 138$
 $= 21661. \text{ Kg}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 82.5) \cdot (6.22 - 3) \cdot 0.7 \cdot 118$$

$$= 7038. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 168 \cdot 10 \cdot 0.74 \cdot 138$$

$$= 27524. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 168 \cdot (9 - 3) \cdot 0.74 \cdot 138$$

$$= 16514. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (21661 + 7038) = 28699 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15584 + 27524 + 16514 + 0) = 59623 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (21661 + 16514 + 0) = 38176 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 28699	Kgf, must exceed W = 0 Kg	or W1 = 14301 Kg
Path 2-2 = 59622	Kgf, must exceed W = 0 Kg	or W2 = 2622 Kg
Path 3-3 = 38175	Kgf, must exceed W = 0 Kg	or W3 = 14767 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 9.38 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 4.1151 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 125.7391 mm.

Input Echo, WRC107/537 Item 1, Description: B18" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	1710.000	mm.
Vessel Thickness	Tv	9.000	mm.
Design Temperature	T1	150.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	3.0000 mm.
Nozzle Diameter	Dn	168.275 mm.
Nozzle Thickness	Tn	6.223 mm.

Nozzle Material	SA-106 B
Nozzle UNS Number	K03006
Nozzle Cold S.I. Allowable	SNmc 118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00 N./mm ²
Thickness of Reinforcing Pad	Tpad 10.000 mm.
Diameter of Reinforcing Pad	Dpad 250.000 mm.
Design Internal Pressure	Dp 3.500 bars
Include Pressure Thrust	No

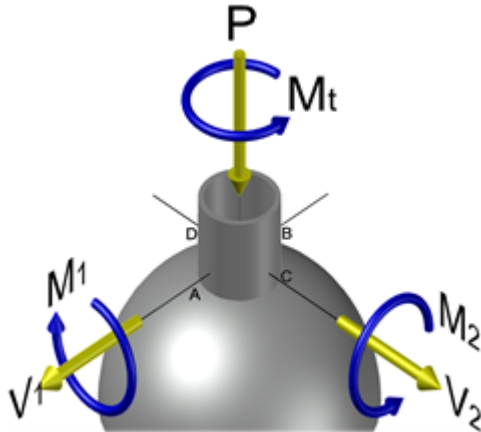
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-1305.3	Kgf
Longitudinal Shear (SUS)	(V1) V1	1153.8	Kgf
Circumferential Shear (SUS)	(Vc) V2	1153.8	Kgf
Circumferential Moment (SUS)	(Mc) M1	605.2	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	605.2	Kg-m.
Torsional Moment (SUS)	Mt	782.9	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 168.275 + 2 \cdot 1.65 \cdot \sqrt{861.0(9.0 - 3.0)} \\
 &= 405.462 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-1305.3	Kgf
Circumferential Shear	(VC) V2	1153.8	Kgf
Longitudinal Shear	(VL) V1	1153.8	Kgf
Circumferential Moment	(MC) M1	605.2	Kg-m.
Longitudinal Moment	(ML) M2	605.2	Kg-m.
Torsional Moment	MT	782.9	Kg-m.

Dimensionless Parameters: U = 0.71 TAU = 25.61 RHO = 4.96

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
----------------------	--------	-------	----------



N(x) * T / P SP 7 0.02666 (A,B,C,D)
 M(x) / P SP 7 0.01243 (A,B,C,D)
 N(x) * T * SQRT(Rm * T) / MC SM 7 0.06008 (A,B,C,D)
 M(x) * SQRT(Rm * T) / MC SM 7 0.02829 (A,B,C,D)
 N(x) * T * SQRT(Rm * T) / ML SM 7 0.06008 (A,B,C,D)
 M(x) * SQRT(Rm * T) / ML SM 7 0.02829 (A,B,C,D)

N(y) * T / P SP 7 0.26781 (A,B,C,D)
 M(y) / P SP 7 0.04671 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / MC SM 7 0.35446 (A,B,C,D)
 M(y) * SQRT(Rm * T) / MC SM 7 0.12645 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / ML SM 7 0.35446 (A,B,C,D)
 M(y) * SQRT(Rm * T) / ML SM 7 0.12645 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Rad. Bend. P		3.7	-3.7	3.7	-3.7	3.7	-3.7	3.7	-3.7
Rad. Memb. MC		0.0	0.0	0.0	0.0	-11.8	-11.8	11.8	11.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-33.4	33.4	33.4	-33.4
Rad. Memb. ML		-11.8	-11.8	11.8	11.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-33.4	33.4	33.4	-33.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-40.2	19.2	50.3	-24.0	-40.2	19.2	50.3	-24.0
Tang. Memb. P		13.4	13.4	13.4	13.4	13.4	13.4	13.4	13.4
Tang. Bend. P		14.0	-14.0	14.0	-14.0	14.0	-14.0	14.0	-14.0
Tang. Memb. MC		0.0	0.0	0.0	0.0	-69.8	-69.8	69.8	69.8
Tang. Bend. MC		0.0	0.0	0.0	0.0	-149.4	149.4	149.4	-149.4
Tang. Memb. ML		-69.8	-69.8	69.8	69.8	0.0	0.0	0.0	0.0
Tang. Bend. ML		-149.4	149.4	149.4	-149.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-191.8	79.0	246.6	-80.2	-191.8	79.0	246.6	-80.2
Shear VC		2.7	2.7	-2.7	-2.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.7	-2.7	2.7	2.7
Shear MT		10.8	10.8	10.8	10.8	10.8	10.8	10.8	10.8
Tot. Shear		13.5	13.5	8.1	8.1	8.1	8.1	13.5	13.5
Str. Int.		193.0	81.9	247.0	81.4	192.3	80.1	247.6	83.3

Unitless Prm: U = 1.74 TAU = 0.00 (38.28) RHO = 0.00 (1.86)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.03049	(A,B,C,D)
M(x) / P	SR 2	0.01468	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.02823	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01939	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.02823	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.01939	(A,B,C,D)
N(y) * T / P	SR 2	0.00914	(A,B,C,D)
M(y) / P	SR 2	0.00441	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00848	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00569	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.00848	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00569	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		10.8	10.8	10.8	10.8	10.8	10.8	10.8	10.8
Rad. Bend. P		31.3	-31.3	31.3	-31.3	31.3	-31.3	31.3	-31.3
Rad. Memb. MC		0.0	0.0	0.0	0.0	-64.7	-64.7	64.7	64.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-266.9	266.9	266.9	-266.9
Rad. Memb. ML		-64.7	-64.7	64.7	64.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-266.9	266.9	266.9	-266.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-289.4	181.6	373.8	-222.6	-289.4	181.6	373.8	-222.6
Tang. Memb. P		3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Tang. Bend. P		9.4	-9.4	9.4	-9.4	9.4	-9.4	9.4	-9.4
Tang. Memb. MC		0.0	0.0	0.0	0.0	-19.4	-19.4	19.4	19.4
Tang. Bend. MC		0.0	0.0	0.0	0.0	-78.3	78.3	78.3	-78.3
Tang. Memb. ML		-19.4	-19.4	19.4	19.4	0.0	0.0	0.0	0.0
Tang. Bend. ML		-78.3	78.3	78.3	-78.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-85.1	52.7	110.4	-65.0	-85.1	52.7	110.4	-65.0
Shear VC		4.8	4.8	-4.8	-4.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.8	-4.8	4.8	4.8
Shear MT		13.0	13.0	13.0	13.0	13.0	13.0	13.0	13.0
Tot. Shear		17.8	17.8	8.2	8.2	8.2	8.2	17.8	17.8
Str. Int.		291.0	184.1	374.0	223.0	289.8	182.2	375.0	224.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		9.4	9.4	9.4	9.4	9.4	9.4	9.4	9.4
Rad. Pl (SUS)		-10.5	-10.5	13.2	13.2	-10.5	-10.5	13.2	13.2
Rad. Q (SUS)		-29.7	29.7	37.2	-37.2	-29.7	29.7	37.2	-37.2
Long. Pm (SUS)		9.4	9.4	9.4	9.4	9.4	9.4	9.4	9.4
Long. Pl (SUS)		-56.4	-56.4	83.2	83.2	-56.4	-56.4	83.2	83.2
Long. Q (SUS)		-135.4	135.4	163.4	-163.4	-135.4	135.4	163.4	-163.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.7	2.7	-2.7	-2.7	-2.7	-2.7	2.7	2.7
Shear Q (SUS)		10.8	10.8	10.8	10.8	10.8	10.8	10.8	10.8
Pm (SUS)		9.4	9.4	9.4	9.4	9.4	9.4	9.4	9.4
Pm+Pl (SUS)		47.2	47.2	92.7	92.7	47.2	47.2	92.7	92.7
Pm+Pl+Q (Total)		183.6	91.3	256.4	72.0	182.9	89.5	257.0	73.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.39	138.00	Passed
Pm+Pl (SUS)	92.69	207.00	Passed
Pm+Pl+Q (TOTAL)	256.95	413.99	Passed



Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0
Rad.	Pl (SUS)	-53.9	-53.9	75.6	75.6	-53.9	-53.9	75.6	75.6
Rad.	Q (SUS)	-235.5	235.5	298.2	-298.2	-235.5	235.5	298.2	-298.2
Long.	Pm (SUS)	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0
Long.	Pl (SUS)	-16.2	-16.2	22.7	22.7	-16.2	-16.2	22.7	22.7
Long.	Q (SUS)	-68.9	68.9	87.7	-87.7	-68.9	68.9	87.7	-87.7
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	4.8	4.8	-4.8	-4.8	-4.8	-4.8	4.8	4.8
Shear	Q (SUS)	13.0	13.0	13.0	13.0	13.0	13.0	13.0	13.0
	Pm (SUS)	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0
	Pm+Pl (SUS)	38.9	38.9	101.0	101.0	38.9	38.9	101.0	101.0
	Pm+Pl+Q (Total)	266.0	209.1	399.1	198.0	264.7	207.2	400.0	199.6

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.03	138.00	Passed
Pm+Pl (SUS)	101.05	207.00	Passed
Pm+Pl+Q (TOTAL)	400.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

Please review all reports carefully!

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: